



Development of a novel physiologically based biopharmaceutics modeling (PBBM) framework using the open systems pharmacology suite, part 2: *in vivo* pharmacokinetic modeling of vericiguat

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ABSTRACT

This manuscript introduces a novel, free and open-source physiologically based biopharmaceutics modeling (PBBM) workflow to support decision-making during oral drug product development. Developed within the Open Systems Pharmacology (OSP) framework, this workflow combines three open-source tools to predict *in vivo* absorption by integrating solubility, dissolution, and systemic pharmacokinetics. The PBBM workflow was successfully applied by 1) deriving solubility and dissolution parameters for vericiguat from *in vitro* experiments as detailed in the accompanying Part 1 publication (Vrenken et al., 2025); 2) then integrating these parameters into a physiologically based pharmacokinetic (PBPK) model. This model incorporates updates to luminal parameters, including pH and bile salt concentration, and employs a new dissolution model accounting for bile salt effects and hydrodynamics. Simulations of vericiguat pharmacokinetics in fasted, fed, and proton pump inhibitor (PPI)-treated states showed good agreement with observed clinical data.

Further refinements that could enhance the PBBM workflow's capabilities were identified. However, the successful application to vericiguat highlights its potential for PBBM applications, including virtual bioequivalence assessments, formulation bridging, and dose selection. While validation across additional case studies is necessary, this free and open-source approach provides a robust foundation for advancing PBBM.

1. Introduction

The oral route of drug administration is widely preferred due to its cost-effectiveness, design versatility, and patient convenience. However, it introduces significant complexity, as drug absorption in the gastrointestinal (GI) tract is influenced by the physicochemical properties of the active pharmaceutical ingredient (API), formulation factors, and the patient's physiological characteristics. Intraluminal physiological factors, such as pH, bile salt concentrations, volumes, and hydrodynamics caused by GI motility can impact solubility and dissolution kinetics *in vivo*. These, in turn, affect the extent to which the API dissolves in the GI tract and becomes available for absorption. Moreover, these

physiological factors can vary within an individual due to events like meal or water intake and co-administration of medications that alter gastric pH (e.g., proton pump inhibitors). They can also vary between individuals due to population differences or disease states such as achlorhydria. Adding to the challenge, many drug candidates in development exhibit poor solubility, increasing the risk of failure due to inconsistent bioavailability (Kambayashi and Dressman, 2022). To address these challenges, oral drug development often relies on animal studies to evaluate the biopharmaceutical behavior of drug candidates and identify robust formulations for human clinical trials (Brake et al., 2017). However, such studies are resource-intensive and time-consuming, and they are further complicated by the poor

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correlation between animal and human bioavailability (Musther et al., 2014). At the same time, there is growing societal demand for more ethical and environmentally sustainable drug development methods, favoring computational models and *in vitro* experiments over animal testing (Reppas et al., 2023).

Physiologically Based Biopharmaceutics Modeling (PBBM) offers an alternative to animal testing by combining absorption modeling, biopredictive dissolution testing, and physiologically based pharmacokinetics (PBPK) to predict systemic drug exposure (Heimbach et al., 2019). PBBM is increasingly being adopted, with applications in bioequivalence assessments, formulation bridging, dose selection, and evaluating factors like food effects and absorption-related drug-drug interactions (DDI) (Mackie et al., 2024). Experience with the application of PBBM remains somewhat limited, particularly in regulatory settings (Manolis et al., 2023; Wu and Li, 2022). However, the U.S. Food and Drug Administration supports the PBBM approach, as reflected in the publication of a draft guidance (FDA, 2020). This method offers mechanistic insights into the *in vivo* absorption process by leveraging the biopharmaceutical properties of the API determined *in vitro*.

In the companion paper, Part 1 (Vrenken et al., 2025), an open-source workflow was developed to inform PBBMs using R (The R Project for Statistical Computing, www.R-project.org) and Open System Pharmacology (OSP) suite based software tools (Lippert et al., 2019). The successful application of these *in silico* tools to assess API solubility and formulation dissolution characteristics using *in vitro* data for vericiguat, a Biopharmaceutics Classification System (BCS) Class 2 compound, was presented. First, the R-Shiny (Chang et al., 2024) Solubility toolbox was employed to assess the impact of pH and bile salt micelles on the *in vitro* solubility of vericiguat. Next, the solubility-related parameters were incorporated into an *in silico* MoBi® dissolution model, which was used to simulate *in vitro* dissolution behavior and to inform formulation specific dissolution properties. All tools are freely available upon publication on OSP GitHub ("OSP: Oral-PBBM-Workflow," 2025).

In this study, we extend on this work by integrating the *in vitro*-derived parameters into an updated PK-Sim® framework for PBPK modeling, incorporating recent advancements in GI physiology modeling. One key area of enhancement is the incorporation of more physiologically relevant luminal GI tract parameters, which govern *in vivo* drug solubility. Based on a comprehensive literature review, these updates include adjustments to luminal pH values and the introduction of population variability, enabling more accurate population-based simulations. Specifically, gastric pH is now dynamic, changing in response to water and food intake, and the upper intestinal pH also adjusts post-meal. Additionally, bile salt concentrations are now implemented and modeled dynamically along the upper GI tract, with increases triggered by food intake. Together, these improvements allow for more accurate modeling of both aqueous solubility (S_{aq}) and apparent solubility enhancements due to bile salt micellization (S_{br}).

The new versatile Noyes-Whitney type dissolution equation, implemented in MoBi®, was also integrated into PK-Sim® (Vrenken et al., 2025). This equation explicitly models two distinct diffusional transports: one for unbound API and another for surfactant-bound API, moving from the dissolving drug particle's surface to the bulk fluid. Additionally, it accounts for hydrodynamic conditions that influence the dissolution process. In PK-Sim®, two extra options have been introduced for modeling the diffusion layer thickness (h), allowing for more flexibility. These options enable users to either keep the diffusion layer thickness equal to the particle radius (with a maximum value) or vary it dynamically based on hydrodynamic conditions, possibly improving the accuracy of dissolution predictions *in vivo*.

This workflow was effectively applied to vericiguat, a novel oral soluble guanylate cyclase stimulator marketed under the tradename Verquvo™ and approved in the United States, the European Union, Japan, China, and various other countries (Markham and Duggan, 2021). Vericiguat is a basic API with a pKa of 4.68 and a LogP of 2.99. It is classified as a BCS Class 2 compound due to its low aqueous solubility

and high permeability (Becker et al., 2022; Frechen et al., 2023). The pharmacokinetics (PK) of vericiguat are affected by prandial state and the use of gastric pH-elevating drugs under fasted conditions. Administering Verquvo™ 10 mg with a high-fat, high-calorie meal reduces PK variability, increasing the area under the curve (AUC_{inf}) by 44 % and the maximum plasma concentration (C_{max}) by 41 % compared to administration in the fasted state (Becker et al., 2022; FDA, 2021). Conversely, pre- or co-administration with gastric pH-elevating drugs, such as proton pump inhibitors or antacids, reduces the AUC_{inf} by approximately 30 % under fasted conditions (Boettcher et al., 2020; FDA, 2021). These properties position vericiguat as an appropriate case study for the newly developed PBBM workflow.

2. Materials & methods

2.1. General overview PBBM workflow

A modular PBBM approach was employed to incorporate solubility and dissolution modeling based on API properties, formulation type, and available *in vitro* data (Vrenken et al., 2025). The software utilized in this novel PBBM workflow is entirely free and open source. For PBPK modeling, the Open Systems Pharmacology (OSP Suite V11.2) tools were used. OSP offers a comprehensive, reliable, open-source, and openly accessible platform for PBPK and quantitative systems pharmacology, specifically through its tools PK-Sim® and MoBi®. More detailed information about OSP can be found in previous multistakeholder reviews (Dallmann et al., 2024; Lippert et al., 2019).

Using the OSP Solubility toolbox (developed with the R-Shiny package, version 1.8.1.1), the aqueous solubility-pH profile and extent of bile salt micelle partitioning was informed from *in vitro* solubility tests. A MoBi® *in silico* dissolution model was used to mimic the *in vitro* dissolution set-up and characterize formulation dissolution kinetics. In the second part of this paper series, a published PK-Sim® PBPK model (Frechen et al., 2023) used for metabolism-related DDI predictions was refined into a PBBM. This was accomplished by integrating the identified solubility and dissolution parameters into the updated PBBM framework. This framework centers around a dissolution model that explicitly models two separate diffusional transports for unbound and surfactant-bound API from the dissolving drug particle's surface to the bulk fluid and accounts for the hydrodynamic conditions. Additionally, the luminal pH and bile salt concentrations were extended and accompanied by population variability to better reflect the increased data reported since the initial implementation (Thelen et al., 2011).

2.2. Parameterization of the PK-Sim® luminal gastrointestinal parameters

A literature review was conducted and the measurements on luminal parameters were collected and included in a Microsoft Excel® (Office 16) spreadsheet. Keywords linked to gastrointestinal fluid characteristics in healthy human adults were combined to conduct a literature search in Google Scholar (<https://scholar.google.com/>). The keywords "human" and "fluid" were combined with a luminal location designation (e.g., stomach/gastric, intestinal, duodenum, etc.) and the parameter of interest (e.g., pH or bile salt/acid concentration). For more granularity, the search was repeated with prandial state designation (e.g., fasted, fed or post-prandial). The collected studies were reviewed to identify additional references that were not recovered during the Google Scholar search.

From the identified studies several details were recorded: (1) the specific parameter, (2) time post meal or water intake in case of temporal data, (3) luminal compartment, (4) water consumed prior to sampling, (5) prandial state, (6) meal type, (7) number of subjects, (8) whether samples were pooled prior to analysis, (9) measurements (mean, SD, percentiles, min, max, individual values), (10) figure or table values were extracted from and (11) DOI. The measurements from

figures were converted to numerical format using WebPlotDigitizer (version 4.7) (Rohatgi, 2024).

Most data were either reported as arithmetic means and standard deviation (SD) or as descriptive statistics rather than individual data, making it difficult to assess the type of distribution in the population. Therefore, unless stated otherwise, it was assumed that the parameters are normally distributed. When multiple mean and SD values were reported the overall mean ($\bar{X}$) and standard deviation ($\overline{SD}$) were calculated according to Eq. (1) and Eq. (2). Where N is the study number, n the number of observations and x the mean in study i .

$$\bar{X} = \frac{\sum_{i=1}^N n_i x_i}{\sum_{i=1}^N n_i} \quad (1)$$

$$\overline{SD} = \sqrt{\frac{1}{\sum_{i=1}^N n_i} \left(\sum_{i=1}^N ((n_i - 1)SD_i^2) + \sum_{i=1}^N (n_i(x_i - \bar{X})^2) \right)} \quad (2)$$

In case only descriptive statistics were reported the mean and SD were calculated according to the methods described Luo et al. and Wan et al., respectively (Luo et al., 2018; Wan et al., 2014). In case of temporally changing luminal parameters following a glass of water or meal, luminal parameter optimization within empirical equations to the temporal literature data was carried out using R (version 4.4.1, The R Project for Statistical Computing, www.R-project.org).

2.3. Gastrointestinal solubility modeling

The solubility along the GI tract was calculated based on the pH and bile salt concentration (C_{BS} , [μM]) within each luminal segment j . Aqueous solubility (S_{aq} , [μM]) was adjusted for ionizable APIs using the PK-Sim® built-in Henderson-Hasselbalch equation, with the solubility gain per charge (SG) setting the maximum attainable S_{aq} . The OSP solubility toolbox was used to inform the SG from *in vitro* solubility experiments, while the intrinsic solubility (S_0 , [μM]) was determined using the API's pKa and the SG (Vrenken et al., 2025). To capture bile salt partitioning, the bile salt micellization tool, paired with solubility data from bile salt media, provided partitioning constants ($K_{m:w,neutral}$ and $K_{m:w,ionized}$) for neutral and ionized API species. Together, the determination of these parameters allow calculation of the apparent or 'biorelevant' (S_{br}) across GI segments using Eq. (3), where S_i ($S_i = S_{aq} - S_0$, [μM]) reflects solubility increase due to ionization, and C_{H2O} represents pure water molarity (55.6e⁶ μM) (Pathak et al., 2017; Sugano, 2009a; Vrenken et al., 2025).

$$S_{br,j}(t) = \left(C_{BS,j}(t) \times \frac{S_0}{C_{H2O}} \times K_{m:w,neutral} + S_0 \right) + \left(C_{BS,j}(t) \times \frac{S_{i,j}(t)}{C_{H2O}} \times K_{m:w,ionized} + S_{i,j}(t) \right) \quad (3)$$

2.4. Gastrointestinal dissolution modeling (in vivo)

An extended dissolution model was integrated into the broader PK-Sim® framework, where solid drug particles travel through the GI tract, including to the rectum, but no dissolution or regrowth occurs in the rectum (Thelen et al., 2012; Willmann et al., 2010). In contrast to the previous implementation, the extended model accounts for micellization. This model (Eq. (4)), based on the film theory, includes two distinct diffusional transports: one for micelle-bound API and one for unbound API, each moving from the particle surface to the bulk fluid in the GI lumen, following the principles outlined by Gamsiz et al. (2010) and Pepin et al. (2019b). These diffusional transports are driven by concentration gradients between the particle surface ($S_{u,surf}$ and $S_{b,surf}$, [μM] for unbound and bound API) and the luminal bulk concentrations ($C_{u,bulk}$ and $C_{b,bulk}$, [μM]). The current model does not account for changes in microenvironmental pH at the solid particle surface compared to the

bulk pH for ionizable APIs. However, the influence of acidic compounds releasing H⁺ ions (lowering pH) and basic compounds consuming H⁺ ions (raising pH), which can reduce API aqueous solubility, can be estimated offline for each intestinal segment using theoretical equations (Uekusa et al., 2021). This surface pH can be estimated for monoprotic acids and bases using the previously presented OSP Solubility Toolbox (Vrenken et al., 2025) across the physiological pH range and the resulting pH adjustments can then be manually applied to the luminal pH values in PK-Sim®, following the approach used by other authors (Hens et al., 2022; Pepin et al., 2019a). It is important to note, however, that in cases where precipitation occurs, the relevant solubility is bulk pH and not surface pH dependent.

$$\frac{dm_{solid,i,j}}{dt} = -A_{i,j}(t) \times \left(\frac{D_u}{h_{u,i,j}(t)} \times (S_{u,surf,j} - C_{u,bulk,j}) + \frac{D_b}{h_{b,j}(t)} \times (S_{b,surf,j} - C_{b,bulk,j}) \right) \times \Psi(t) \quad (4)$$

The dissolution rate depends on the available particle surface area (A , [cm²], Eq. (5)) in a specific GI segment, which decreases over time as dissolution progresses. Assuming spherical particles, the model divides them into particle size bins (i), where each bin in a segment contains a certain number of particles (N). Dissolution within each luminal segment (j) is tracked until particle radii (r_i) in the bins reduce below a threshold (1e⁻⁵ cm), at which point they are considered fully dissolved.

$$A_{i,j}(t) = N_{i,j}(t) \times 4 \times \pi \times r_{i,j}^2(t) \quad (5)$$

The diffusion coefficient of the unbound API (D_u , [cm²/min]) was calculated from the molecular weight based on molecular weight using PK-Sim®'s default empirical equation (Avdeef et al., 2004). For the bound API, the diffusion coefficient (D_b , [cm²/min]) was assumed to match the micelle diffusion coefficient, calculated via the Stokes-Einstein equation (Eq. (6)) using the micelle radius (r_m , [cm]) (Okazaki et al., 2008; Pepin et al., 2019b). The micelle radius was set to the value from fasted-state simulating intestinal fluid ($r_{m,FaSSIF} = 2.72e^{-6}$ cm) (Okazaki et al., 2008), acknowledging that *in vivo* conditions may vary due to the presence of diverse micelle structures with fluctuating sizes and changing over time (Mullertz et al., 2015). The micelle radius is influenced by factors such as bile salt concentration, the ratio of different bile salts, and phospholipid concentrations (Sugano et al., 2007). In this equation, k_b represents the Boltzmann constant, T the absolute temperature and η the dynamic viscosity is (in [J/K], [K] and [Pa*s], respectively).

$$D_b = \frac{k_b \times T}{6 \times \pi \times \eta \times r_m} \quad (6)$$

Diffusion layer thickness is different for the bound and unbound API (h_u and h_b , [cm]), and is influenced by the hydrodynamic conditions surrounding dissolving drug particles (e.g., fluid velocity and viscosity). *In vivo* these conditions are vastly different than in dissolution devices, e.g., the United States Pharmacopeia 2 (USP2). Therefore three options were implemented in PK-Sim®, i) a constant value (30 μm, default), ii) the empirical Hintz-Johnson model where $h = r_t$ unless the particle radius is over 30 μm (adjustable) then $h = 30$ μm, iii) a simplified theoretical model that relates h in addition to r_t to *in vivo* luminal hydrodynamics.

The (semi-)mechanistic model extensively described by Pepin et al., 2022 and implemented in MoBi® for modeling h_u and h_b during USP2 dissolution experiments was simplified to model the complex *in vivo* luminal hydrodynamics (Pepin et al., 2022; Vrenken et al., 2025). The simplified hydrodynamic model considers the intestinal segments as a smooth tube where particles move synchronously with the fluid (e.g., neglecting intestinal folding and peristaltic movements) (Sugano, 2010). Given the relatively low fluid flow velocities, small intestinal, and high

intestinal fluid viscosity, the intestinal fluid flow was assumed to be laminar. In line with Computational fluid dynamics (CFD) studies showing consensus that gastric and intestinal flows are primarily laminar rather than turbulent (Li and Kong, 2022). h_u was calculated from the Sherwood number (Sh) and r_t (Eq. (7)). h_b in turn, can be calculated from h_u with Eq. (8) (Pohl et al., 1998).

$$h_{u,ij}(t) = \frac{2 \times r_{ij}(t)}{Sh} \quad (7)$$

$$h_{b,ij}(t) = h_{u,ij}(t) \times \sqrt[3]{\frac{D_b}{D_u}} \quad (8)$$

Sh can be estimated using the Ranz-Marshall equation, which incorporates the particle Reynolds number (Re) and the Schmidt number (Sc), as depicted in Eq. (9) for spherical particles.

$$Sh_{ij} = 2 + 0.6Re_{ij}^{1/2}(t) \times Sc_j^{1/3} \quad (9)$$

Re represents the ratio of inertial to viscous forces in fluid flow and is key to understanding intraluminal flow patterns and can be calculated from fluid kinematic viscosity (ν , [cm²/min]), average fluid velocity ($\bar{V}_{fluid}$, [cm/min]) and r_t (Eq. (10)). ν in the stomach and intestine were set to measured values in fasted state of 2.0×10^{-2} and 4.7×10^{-1} cm²/min, respectively (Litou et al., 2016; Rivera et al., 2023). $\bar{V}_{fluid}$ was related to the luminal liquid transfer rate (Q , [L/min]) and the average radius of the luminal segment ($\bar{r}_{lum}$, [cm]) through Eq. (11), assuming the dissolving particles move synchronously with the intestinal fluid. For the gastric compartment a constant average value was set taken from CFD simulations (28.5 cm/min, (Li and Kong, 2022)). Sc was calculated with Eq. (12).

$$Re_{ij}(t) = \frac{2r_{ij}(t) \times \bar{V}_{fluid,j}(t)}{\nu_j} \quad (10)$$

$$\bar{V}_{fluid,j}(t) = \frac{Q_j(t)}{\pi \bar{r}_{lum,j}} \quad (11)$$

$$Sc_j = \frac{\nu_j}{D_u} \quad (12)$$

The final parameter in Eq. (4) is the empirical ‘absolute’ surface integration factor (Ψ , [cm], Eq. (13)), which serves to reduce particle growth rate when concentrations exceed API solubility (Lindfors et al., 2008; Sjögren et al., 2013). This factor adjusts for surface processes that slow particle growth and deviate from Fick’s first law of diffusion. During dissolution this factor is set to 1 and has no influence on the dissolution rate. The API or formulation specific surface integration factor (ϵ , [cm]), can be estimated from multi-stage *in vitro* dissolution tests or clinical data when precipitation is observed (Kostewicz et al., 2004; Kourentas et al., 2016; Mann et al., 2017). Setting ϵ to a high value can nearly eliminate particle growth altogether.

$$\Psi(t) = \begin{cases} 1, & \text{if } \frac{dm_{solid,j}}{dt} \geq 0 \\ \frac{r_{ij}(t)}{r_{ij}(t) + \epsilon}, & \text{otherwise} \end{cases} \quad (13)$$

2.5. PBBM approach integrating solubility and dissolution into the vericiguat PBPK model

The models described above were integrated into a vericiguat PBPK model, that has been previously qualified to be used for uridine diphosphate-glucuronosyltransferase (UGT) mediated DDIs (Frechen et al., 2023), in order to create a PBBM. This PBPK model (termed ‘original’ model) describes both the vericiguat and primary metabolite (M1) distribution, metabolism, and excretion. Several phase-1 studies including single ascending dose with oral solutions, intravenous

micro-tracer, and radiolabeled oral solution administration were used to inform and qualify the original model.

Several key processes from the original PBPK model were left unchanged during PBBM development. These are vericiguat metabolism into M1 by UGT1A9 (in kidneys and liver) and UGT1A1 (mainly in the liver), vericiguat oxidation by CYP1A1 and CYP3A4 to a lesser extent, active vericiguat transport via P-gp and BCRP, cleavage of M1 back to vericiguat following M1 biliary clearance (presumably by beta-glucuronidases present in intestinal microbiota) and finally, kidney tubular excretion of M1 in addition to glomerular filtration of both vericiguat and M1 (Frechen et al., 2023).

In this study, the mechanistic dissolution processes described in Sections 2.3. and 2.4. were newly incorporated and parametrized into the original PBPK model, enabling a bottom-up approach. This enhancement allows for extrapolation to untested clinical scenarios that could affect drug absorption, capabilities that were not feasible with the previous PBPK modeling approach and were not required for its intended use (Frechen et al., 2023). However, unlike the previously published model, the presented PBBM was not applied to predict metabolic DDIs and involves a few differences in parameterization discussed in the following. Vericiguat intestinal solubility was described by Eq. (3) where the parameters SG and $K_{m,w}$ values were inferred from measured *in vitro* solubilities. Due to the new model implementation of solubility and parameterization, refitting of the intestinal permeability for the PBBM was required. Luminal bile salt concentrations and updated luminal pH values were implemented into the PK-Sim® software. A literature search was conducted to determine relevant luminal parameters and its assigned population variability.

Tablet dissolution in the original model was described using Weibull functions optimized to the clinical data, which were replaced in this work with formulation specific product-particle size distributions (P-PSD) (Pepin et al., 2016). The P-PSD was used instead of the measured API powder particle sizes because the latter may not accurately represent the surface area available for dissolution in the final formulation. Factors such as particle clumping, irregular shapes, and excipients affecting wettability can alter dissolution kinetics in the final product (Pepin et al., 2022, 2016). The P-PSDs for the tablets were informed from *in vitro* dissolution experiments as described in part 1 of this paper series (Vrenken et al., 2025). Based on internal data (not shown) from two-stage dissolution assays using fasted-state simulated gastric fluid (FaSSGF, pH 1.6) followed by the addition of concentrated fasted-state simulated intestinal fluid (FaSSIF, pH ~6.5) across the 5–15 mg dose range, where no precipitation was observed upon transition to intestinal conditions, the specific surface integration factor (ϵ) was set to a very high value to inhibit precipitation in the PBBM. An overview of the clinical studies used for developing and evaluating the final PBBM is summarized in Table 1, where the formulation that was used as P-PSD input in the present PBBM approach is indicated in the respective column. All investigated vericiguat tablets were similar crystalline immediate release (IR) formulations but differed in dose strength, the P-PSD used in the simulations for each tablet formulation is summarized in Table S1. The PBBM model files are available on OSP GitHub.

2.6. PBBM evaluation

The studies listed in Table 1 where tablets were administered in the fasted and fed state were utilized to evaluate the PBBM performance. A virtual cohort of 1000 healthy European volunteers, matched to the demographic criteria outlined in the clinical studies of vericiguat was created to run the population simulations: male participants aged between 18 and 55 years, with a body mass index ranging from 18 to 30 kg/m². Only for the vericiguat PPI DDI clinical study arms separate populations were created for both the treatment and control group matching the enrolled participants demographic parameters (Boettcher et al., 2020). To resemble the treated population following a 5-day PPI omeprazole pretreatment of 40 mg once-daily, basal stomach pH in

Table 1

Summary of the clinical studies used for PBBM development and evaluation.

ID	Clinical studies used for developing and evaluating the PBBM (Trial ID) ^a	n	Dose [mg]	Formulation (P-PSD input)	Food status ^b	Data used for
A	Single ascending dose (15355)	56	0.5–15	Solution	Fasted	PBBM development
B	Relative bioavailability (15356)	15	1.25	1.25 mg tablet	Fasted	PBBM evaluation
		15	5	5 mg tablet	Fasted	
		15	5	Solution	Fasted	PBBM development
C	Multiple ascending dose, first 24 h only (15357)	9	1.25	1.25 mg tablet	Fasted	PBBM evaluation
		8	5	1.25 mg tablet	Fasted	
		8	10	1.25 mg tablet	Fasted	
D	Mass balance (15817)	6	5	Solution	Fasted	PBBM development
E	Aspirin, pilot, and control phase (15838)	25	15	5 mg tablet	Fasted	PBBM evaluation
F	Relative bioavailability (16440)	16	1.25	1.25 mg tablet	Fed	PBBM evaluation
		15	2.5	1.25 mg tablet	Fed	
		14	5	5 mg tablet	Fed	
		16	10	5 mg tablet	Fed	
G	Pivotal food effect study (18580)	29	2.5	2.5 mg tablet	Fed	PBBM evaluation
		30	5	5 mg tablet	Fed	
		29	10	10 mg tablet	Fed	
		29	10	10 mg tablet	Fasted	
H	PPI DDI, control phase (15811)	10	5	1.25 mg tablet	Fasted	PBBM evaluation
	PPI DDI, omeprazole (15811)	10	5	1.25 mg tablet	Fasted	

Abbreviations: n = Number of study participants, PPI = Proton pump inhibitor, DDI = Drug-drug interaction, P-PSD = Product-Particle Size Distribution.

^a Bayer trial identifier, can be used to access study details in the Bayer Clinical Trials Explorer ("Bayer - Clinical Trials Explorer," 2025).^b Fed indicates vericiguat was administered following a standard high-fat, high-caloric breakfast.

fasted state and standard deviation were adjusted to 4.6 ± 1.4 according to literature data collected after similar PPI treatments for the treated group (Kirchheiner et al., 2009). Predicted plasma concentration-time profiles were used to derive the PK parameters such as area under the curve (AUC), maximal plasma concentration ($C_{\max}$) and the time of $C_{\max}$ ($t_{\max}$). Performance of the PBBM was assessed by its ability to predict these PK parameters. Predictive performance was visibly assessed using dose normalized plots over the investigated dose range and quantified using the metrics fold error ($FE = \text{pred}_i/\text{obs}_i$), average fold error (AFE, Eq. (14)) and absolute average fold error (AAFE, Eq. (15)), where obs_i is the observed and pred_i the predicted PK parameter value. AFE and FE assess bias, with values <1 indicating overprediction and >1 indicating underprediction; AFE or FE = 1 reflects perfect agreement. AAFE captures the magnitude of prediction error, treating under- and over-predictions symmetrically via the absolute log fold error. Performance was classified as good (AFE 0.8–1.25; AAFE ≤ 1.25), acceptable (AFE 0.5–0.8 or 1.25–2; AAFE 1.25–2), or poor (AFE <0.5 or >2 ; AAFE >2). The AFE and AAFE were applied to assess the overall performance for the fasted and fed state separately.

$$AFE = 10^n \sum \log \frac{\text{pred}_i}{\text{obs}_i} \quad (14)$$

$$AAFE = 10 \sum \left| \log \frac{\text{pred}_i}{\text{obs}_i} \right| \quad (15)$$

The model's sensitivity to variations in absorption-related parameters was investigated, focusing on their impact on $C_{\max}$ and AUC. These absorption-related parameters were identified during the *in vitro* steps of the PBBM workflow as described before and through the literature study on luminal parameters described above (Vrenken et al., 2025). Sensitivity testing was conducted by adjusting each parameter independently over a broad range, up to three orders of magnitude above and below the baseline. All plasma concentration-time, dose normalized, and sensitivity plots were created in R.

3. Results

3.1. Adjustment of luminal gastrointestinal parameters

3.1.1. Luminal pH values, bile salt concentrations hydrodynamic parameters

Table 2 provides an overview comparing the newly implemented

Table 2

Overview implemented fasted state luminal pH values and bile salt concentrations [mM].

Lumen segment	No. of Studies (n) ^a	pH (SD)	Previous PK-Sim pH	No. of Studies (n) ^a	Bile salt conc. (SD) [mM]
Stomach (without water)	5 (61)	1.82 (0.51)	2	1 (36)	0.19 (1.2) ^b
Duodenum	9 (103)	6.27 (0.80)	6	9 (122)	3.6 (3.27)
Upper jejunum	9 (162)	6.69 (0.73)	6.25	6 (73)	2.69 (2.46)
Lower jejunum ^c	9 (162)	6.69 (0.73)	6.92	6 (73)	2.69 (2.46)
Upper ileum	1 (48)	7.49 (0.46)	7.21	–	0.071 (0.15) ^d
Lower ileum	1 (12)	8.08 (0.52)	7.46	1 (12)	0.071 (0.15)
Caecum	1 (12)	7.36 (0.63)	5.7	1 (12)	0.182 (0.22)
Ascending colon	1 (20)	7.66 (0.56)	5.6	1 (20)	0.155 (0.12)
Transverse colon	1 (48)	6.6 (0.59)	5.7	0 ^e	0 ^e
Descending colon	1 (48)	6.6 (0.59)	6.6	0 ^e	0 ^e
Sigmoid colon	1 (48)	6.6 (0.59)	6.6	0 ^e	0 ^e
Rectum	1 (48)	6.6 (0.59)	6.6	0 ^e	0 ^e

Abbreviations: n = Number of study participants, SD = Arithmetic standard deviation.

^a Individual references and values are depicted in Table S2 and Table S3.^b Geometric mean and standard deviation instead of arithmetic values, (Lindahl et al., 1997).^c Lower jejunal pH and bile salt concentration assumed to be similar to the upper jejunum, due to absence of data reported.^d Assumed to be similar to the lower ileum, due to absence of data reported.^e Set to zero, due to absence of data reported.

mean luminal pH values with the previous values. In the new implementation, population variability has also been incorporated for both pH and mean bile salt concentrations. All included literature studies regarding the fasted luminal pH and bile salt concentrations are shown in Table S2 and Table S3. Stomach pH values were only included when

the pH was determined immediately upon collection, to prevent sample pH drift over time due to buffer capacity changes (Litou et al., 2016). Additionally, pooled samples were not included for analysis of intestinal pH values.

3.1.2. Dynamic gastrointestinal pH

Stomach pH increases temporarily ($pH_{Sto,t}$) following a glass of water and should be considered (Koenigsknecht et al., 2017; Litou et al., 2016; Petrakis et al., 2015). In the new PK-Sim implementation the hydrogen ion concentration was assumed to be constant, while the administered water leads to dilution and a rise in stomach pH. This dilution effect was described by Eq. (16), where stomach pH returns to baseline as soon as the liquid volume ($V_{Sto,t}$) returns to the basal volume ($V_{Sto,Basal}$). The ability of the gastric fluid dilution model to describe the stomach pH is depicted in Fig. 1A, where the shaded area describes the 95 % confidence interval of the basal stomach pH (pH_{Basal}).

$$pH_{Sto}(t) = -\text{Log}_{10}\left(10^{[-pH_{Basal}]} * \frac{V_{Sto,Basal}}{V_{Sto}(t)}\right) \quad (16)$$

A similar approach for modeling pH following a high-fat, high-caloric meal was unsuccessful when based solely on the stomach volume. This was likely due to the buffering effect of food constituents, which resulted in a larger pH increase than expected. Additionally, the return to basal gastric pH was slower than observed (Fig. 1B). To better reflect post-meal gastric pH dynamics, the inbuilt gastric pH decay model (Eq. (17)) was refitted to observed data (Dressman et al., 1990; Koziol et al., 2015; Pentafragka et al., 2020). In this model, the decay constant α was optimized ($\alpha = 1.72e^{-2}$), with pH_{Fed} representing the initial pH following the meal (set to 4.85), and t_{Meal} is the time of meal administration. The refitted post-meal decay model adequately captured the observed data, as shown in Fig. 1B.

$$pH_{Sto}(t) = pH_{Basal} + (pH_{Fed} - pH_{Basal}) \times e^{[-\alpha(t-t_{Meal})]} \quad (17)$$

Following a glass of water the upper intestinal pH was kept constant after a high fat-high calorie meal the duodenal and upper jejunal decrease (Pentafragka et al., 2020). This decrease was assumed to be similar in these luminal segments and a 2nd order polynomial was used to describe the reduction in pH and return to fasted pH (Eq. (18)).

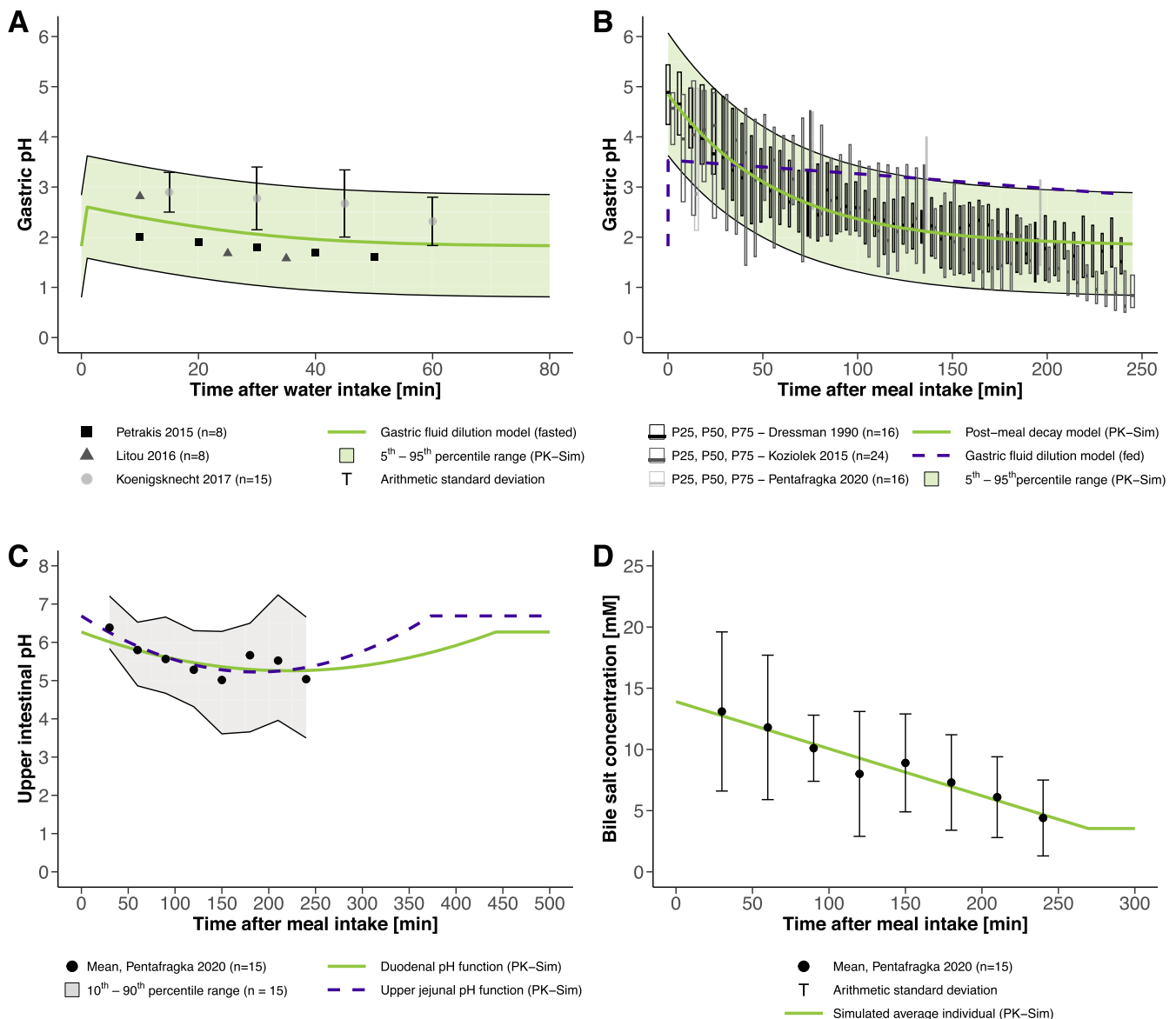


Fig. 1. Comparison of observed against dynamically modeled luminal parameters over time. Panel A depicts the gastric pH following a glass of water in the fasted state and panel B the gastric pH after a high-calorie, high-fat meal. Panels C and D depict the upper intestinal pH and bile salt concentration, in mM, in the fed state.

$$pH_{Duo/U.jej}(t) = c1 \times t^2 - c2 * t + pH_{Fasted} \quad (18)$$

The coefficients, $c1$ and $c2$, were optimized separately for both luminal segments, the description of the luminal pH following a meal in the duodenum ($c1 = 2.07e^{-5}$, $c2 = 9.16e^{-3}$) and upper jejunum ($c1 =$

$4.21e^{-5}$, $c2 = 1.57e^{-2}$) can be seen in Fig. 1C.

3.1.3. Luminal bile salt concentrations

Studies on fasted state bile salt concentrations were only included if

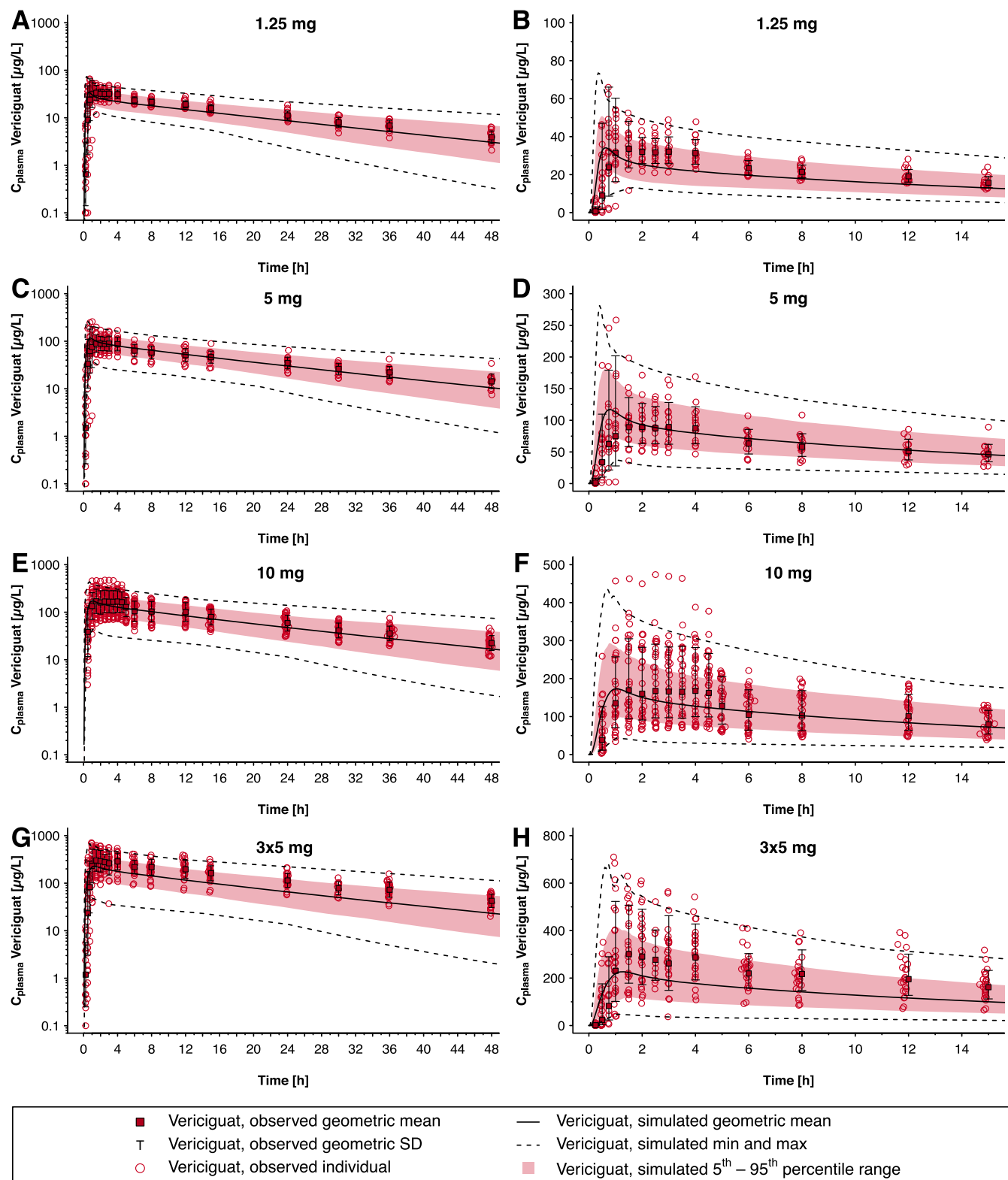


Fig. 2. Simulated *versus* observed concentration-time profiles following application of 1.25 mg (A, B), 5 mg (C, D), 10 mg (E, F) or 3 × 5 mg (G, H) vericiguat tablet (s) in fasted state. The left panels display the first 48 hours with vericiguat concentration on the y-axis using a log scale, right panels show the first 15 hours with vericiguat concentration on the y-axis using a linear scale.

they did not pool samples prior to measurement. In case multiple measurements at different time points were reported they were combined to a single mean and SD value using Eq. (1) and Eq. (2). The final mean value, SD value and reported study values can be seen in Table 2.

Following a high-fat, high-caloric meal the return to fasted state bile salt concentration ($C_{BS,Fasted}$, [μM]) was well described by a linear function (Eq. (19)).

$$C_{BS,j}(t) = C_{BS,Fed} - a * t, C_{BS,Fasted,j} \tag{19}$$

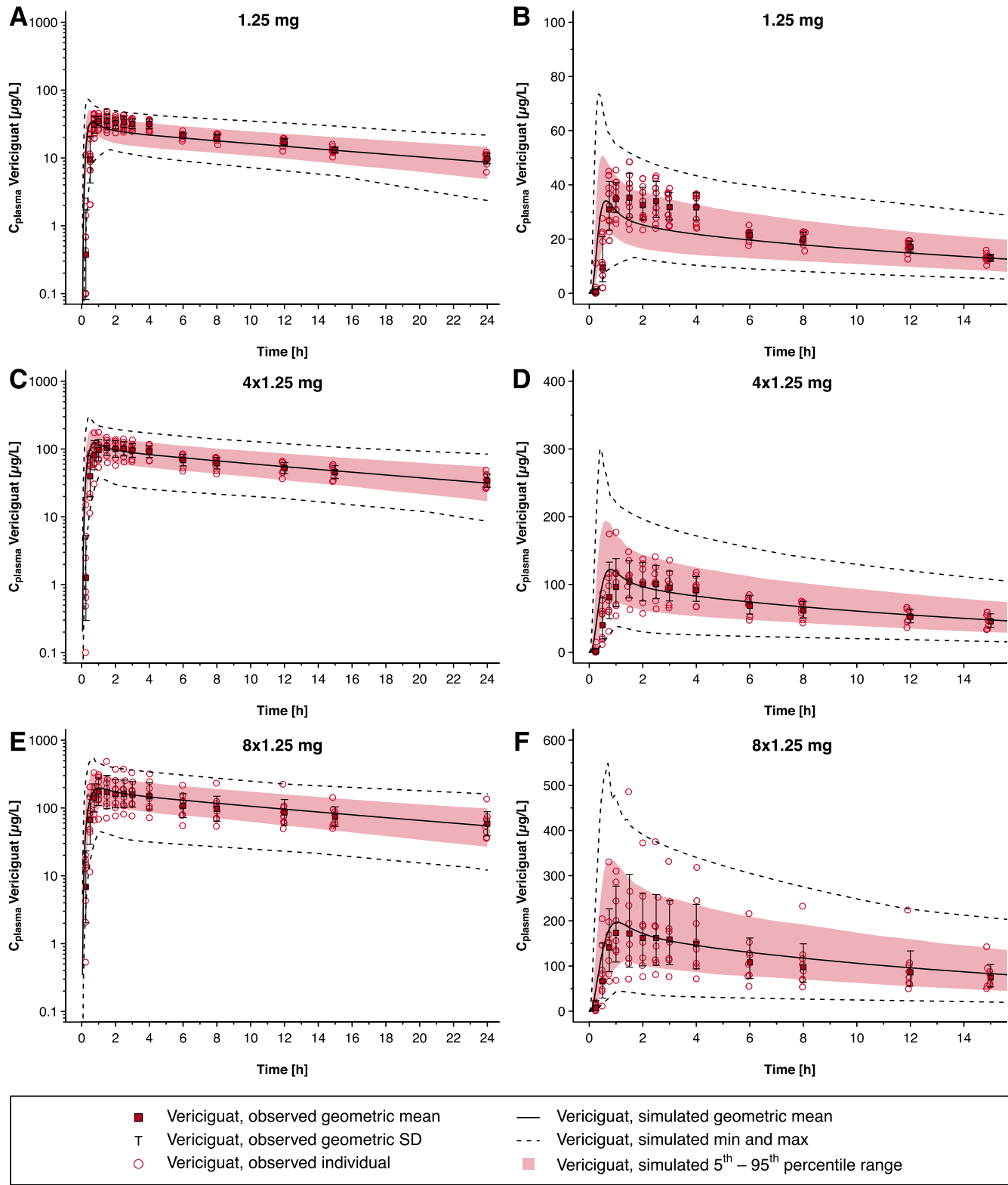


Fig. 3. Simulated versus observed concentration-time profiles following 1.25 mg (A, B), 5 mg (C, D) and 10 mg (E, F) as one or multiple 1.25 mg tablets in fasted state. The left panels display the first 24 hours with vericiguat concentration on the y-axis using a log scale, right panels show the first 15 hours with vericiguat concentration on the y-axis using a linear scale.

Where the fed state bile salt concentration ($C_{BS,Fed} \pm SD = 13,904 \pm 6500$ [μM]) and a , the bile salt slope factor ($a = 3.71e^{-3}$ [min^{-1}]), were inferred from observed data as shown in Fig. 1D (Pentafragka et al., 2020). C_{BS} was set to $C_{BS,Fasted}$ once it became lower than this value.

3.2. PBBM results

3.2.1. Reparameterization of intestinal permeability

The intestinal permeability of the vericiguat PBBM was refitted, because of the solubility related changes to the model and vericiguat parameters, using the PK-Sim® parameter identification tool. Although, the exposure following a 15 mg oral solution ($n = 4$) was somewhat underpredicted, the final model provided an adequate description of the observed plasma PK data following an oral solution over a dose range of 0.5–10 mg (Figure S1). After the model reparameterization, it was considered suitable to be used as PBBM for simulating vericiguat exposure after tablet administration. An overview of the final PBBM input parameters can be found in Table S4.

3.2.2. Fasted state simulations

The *in vitro* dissolution-informed P-PSDs were incorporated into the PBBM as formulation and administration building blocks in PK-Sim® (Table S1). This model accurately predicted clinically observed vericiguat plasma concentration-time profiles after administration of one or multiple tablets in a dose range of 1.25–15 mg in a fasted state with 240 mL of water (Fig. 2). The model also satisfactorily described vericiguat PK following single and multiple administrations of the 1.25 mg tablets at 5 mg and 10 mg doses (Fig. 3).

Predictions for key PK parameters such as C_{max} and AUC were close to values observed in clinical studies the AFE (C_{max} : 1.00, AUC: 0.88) and AAFE (C_{max} : 1.17, AUC: 1.19), indicating good predictive performance (Fig. 4). However, the model underpredicted AUC_{inf} (FE: 1.64) and C_{max} (FE: 1.46) following the highest dose of 15 mg, these predictions were deemed acceptable. The predicted median t_{max} fell within the observed range but was generally on the low end of the observed range in the fasted state (AFE: 0.54, AAFE: 1.85). An overview of the observed and predicted PK parameter, FE, AFE and AAFE values in the fasted and fed state is shown in Table S5.

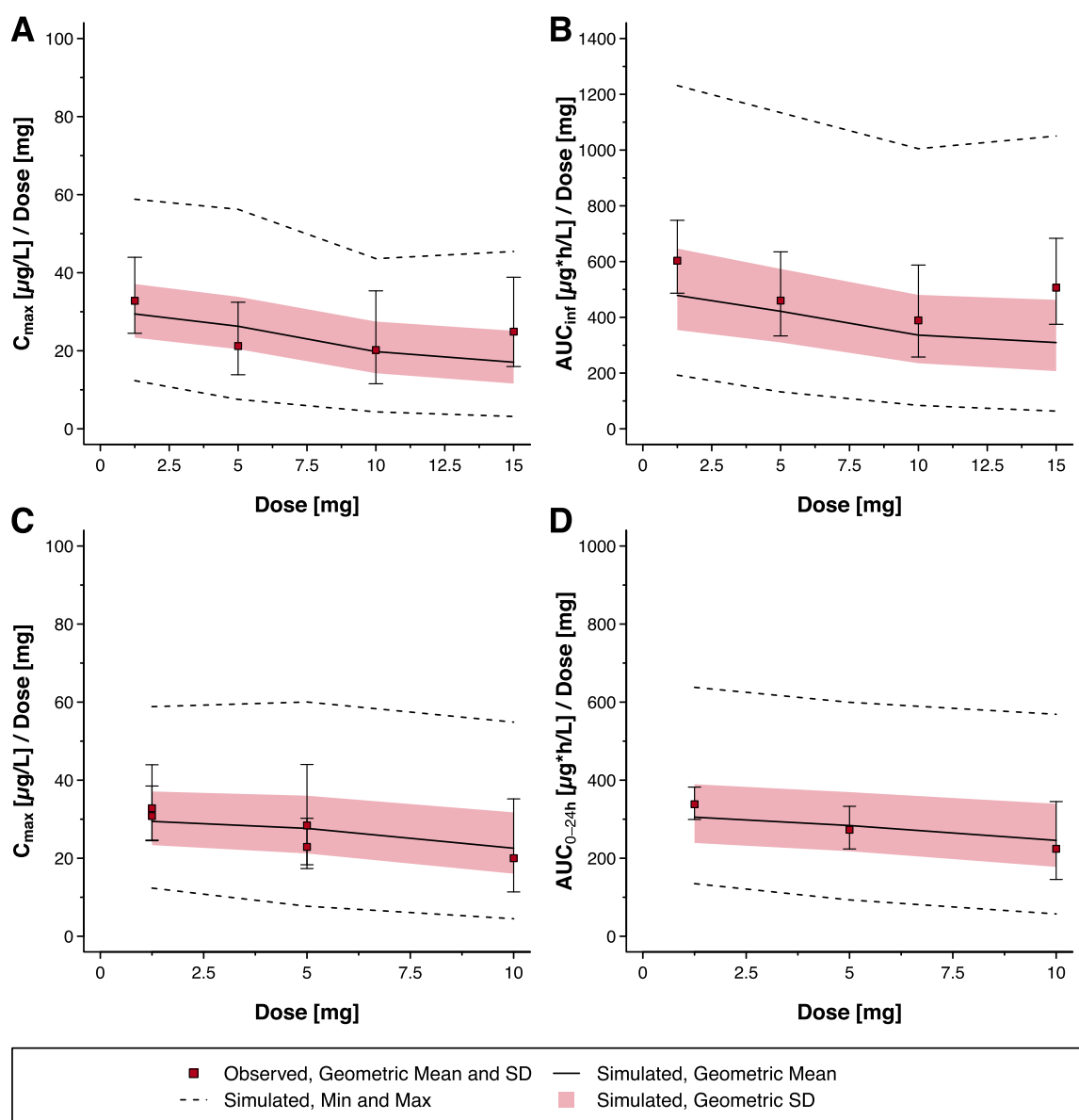


Fig. 4. Simulated *versus* observed dose-normalized pharmacokinetic parameters C_{max} (A, C), AUC_{inf} (B) and $AUC_{0-24 h}$ (D) in fasted state, the bottom panels C and D contain the data following one or multiple 1.25 mg vericiguat tablet(s).

3.2.3. PPI interaction simulation

The model effectively captured the impact of repeated omeprazole administration on vericiguat's PK in the fasted state. The observed reductions in $C_{\max}$ (−49 %) and AUC_{inf} (−32 %) geometric mean following omeprazole administration were closely aligned with the model predictions for $C_{\max}$ (−47 %) and AUC_{inf} (−39 %), confirming the model's ability to replicate the PPI interaction effects (Boettcher et al., 2020). Despite this, $t_{\max}$ was underpredicted both with and without omeprazole (Fig. 5).

3.2.4. Fed state simulations

Using the newly implemented dynamic physiological models for the fed state, the effect of food on the absorption of vericiguat was reasonably well captured, without the need for additional parameter adjustments. The model accurately predicted the delayed $t_{\max}$ (AFE: 1.17, AAFE: 1.17, Table S5) in the fed state compared to the fasted state, reflecting the food-induced delay in absorption (Fig. 6 & Fig. 7). Additionally, AUC_{inf} predictions in the fed state were in good agreement with observed values across the 1.25–10 mg dose range (AFE: 0.86, AAFE: 1.17). The predicted increase of AUC_{inf} of a 10 mg tablet following a high-fat high-calorie meal compared to the fasted state was 38 %, which is close to the observed increase of 41 % in the clinic. However, $C_{\max}$ was underpredicted in the fed state (AFE: 0.66), indicating a deviation in peak concentration predictions (Fig. 8). The overall fed $C_{\max}$ prediction was deemed acceptable as indicated by an AAFE of 1.51.

3.2.5. Sensitivity analysis

The sensitivity analysis indicates that vericiguat's intrinsic solubility is the most influential parameter affecting $C_{\max}$ and AUC in both the fed and fasted states (Fig. 9). In the fasted state, a ± 1 log fold change in intrinsic solubility resulted in a ± 56 %/−85 % change in $C_{\max}$ and a ± 48 %/−81 % change in AUC. In the fed state, the corresponding impact was reduced, with changes of ± 16 %/−38 % in $C_{\max}$ and ± 19 %/−39 % in AUC. Bile salt partitioning exhibited greater sensitivity in the fasted state, producing changes of ± 8 %/−16 % in $C_{\max}$ and ± 4 %/−7 % in AUC. In contrast, in the fed state, changes in $C_{\max}$ and AUC were negligible (≤ 1 %), with larger increases beyond 1 log unit required to observe any significant effect. Additionally, gastric hydrodynamics ($\bar{V}_{\text{fluid}}$) influenced PK parameters, with changes in $C_{\max}$ and AUC ranging

from 4 % to 8 % in both the fasted and fed states. Whereas, upper intestinal hydrodynamics had no significant impact on the simulation outcomes for vericiguat. The SIF affected the $C_{\max}$ and AUC only in the fasted state after a 2 or higher log fold decrease.

4. Discussion

In this work, we describe the final step in a novel open-source PBBM workflow: integrating *in vitro*-informed solubility and dissolution parameters of poorly soluble APIs into an updated PK-Sim® PBBM. This integration includes refinement of gastrointestinal parameters and the dissolution model. The approach was successfully applied to capture the PK of vericiguat in both fasted and fed states, as well as in the fasted state following PPI pre-treatment. *In vitro* dissolution data were mechanistically incorporated into the PBBM by fitting a P-PSD that aligned with the observed dissolution kinetics, enabling effective “bottom-up” predictions. This workflow has potential future applications for *in silico* bioequivalence assessments, formulation bridging, and dose selection.

4.1. Adjustment of luminal gastrointestinal parameters

The updated PK-Sim® framework extends its existing GI parameters by integrating revised luminal pH values and newly introduced bile salt concentrations, both now accounting for population variability. These enhancements improve the model's ability to capture individual differences in gastrointestinal conditions that influence drug absorption, thereby strengthening its application in population-based PK analyses. To maximize the number of studies included, luminal bile salt concentrations were modeled using a normal distribution with a lower cut-off set at 0 μM . While these concentrations may follow a different distribution in reality (e.g., log-normal distribution), the available literature data did not allow for a more accurate assessment of the true distribution.

For simulating gastric pH dynamics after drinking a glass of water, a gastric fluid dilution model was used, which assumes a constant amount of hydrogen ions in the stomach. This model successfully captured observed data following water intake, providing an effective way to represent pH changes under common dosing conditions (Fig. 1A). This simplified model allows for simulations under real-world dosing

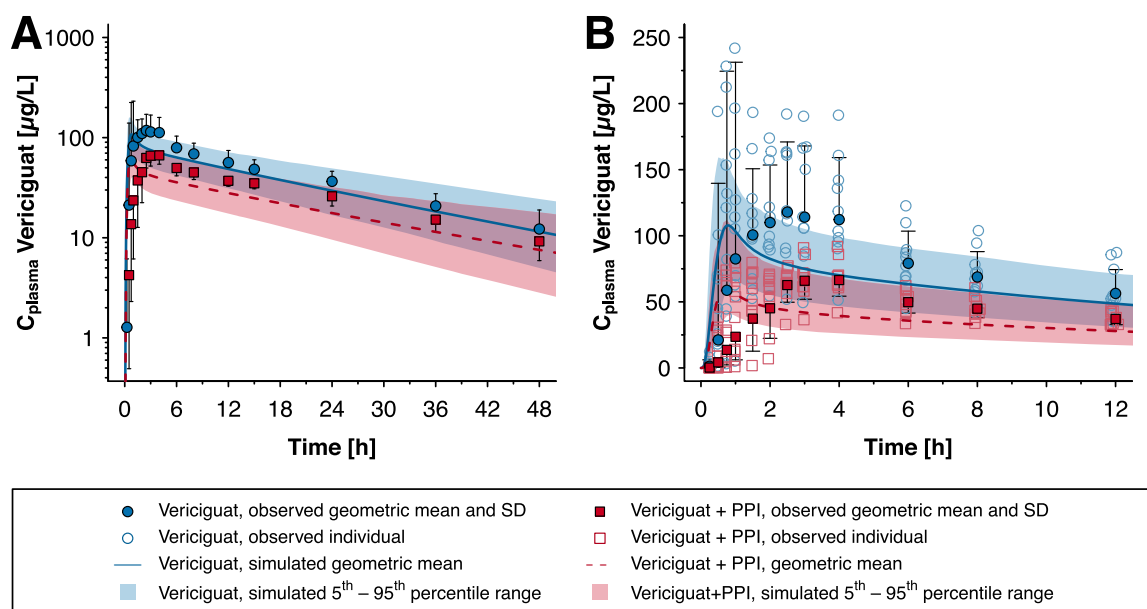


Fig. 5. Simulated versus observed plasma concentration-time profile after dosing of 5 mg vericiguat with and without pre-treatment of proton-pump inhibitor omeprazole. The left panel depicts the first 48 h with the concentrations on the y-axis using a log scale (A) and the right panel shows the first 12 h using a linear y-scale (B).

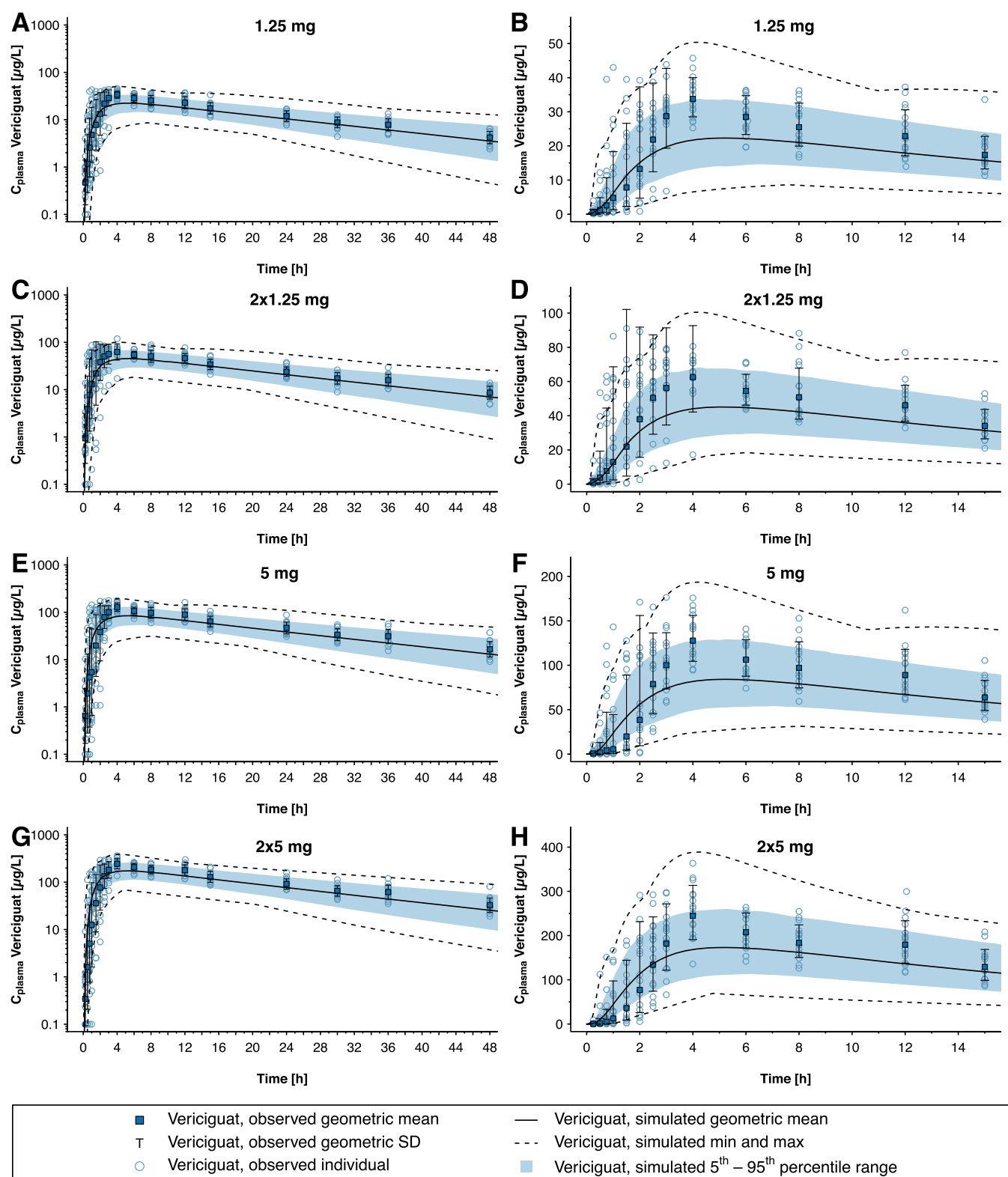


Fig. 6. Simulated *versus* observed plasma concentration-time profiles after dosing of 1.25 mg (A, B), 2×1.25 mg (C, D), 5 mg (E, F) or 2×5 mg (G, H) vericiguat tablet(s) following a high-fat, high-caloric meal. The left panels display the first 48 hours with vericiguat concentration on the y-axis using a log scale, right panels show the first 15 hours with vericiguat concentration on the y-axis using a linear scale.

conditions where smaller volumes of water are often taken during oral drug administration, enhancing its practical applicability (Hummeler et al., 2023). However, it is important to acknowledge that variations in pH distribution throughout the stomach may lead to inconsistencies,

particularly in the fed state, where pH heterogeneity can be more pronounced (Simonian et al., 2005). Several dynamic factors influence gastric pH, such as fluctuating rates of acid secretion, saliva production, stomach emptying, and duodenal reflux, all of which can introduce

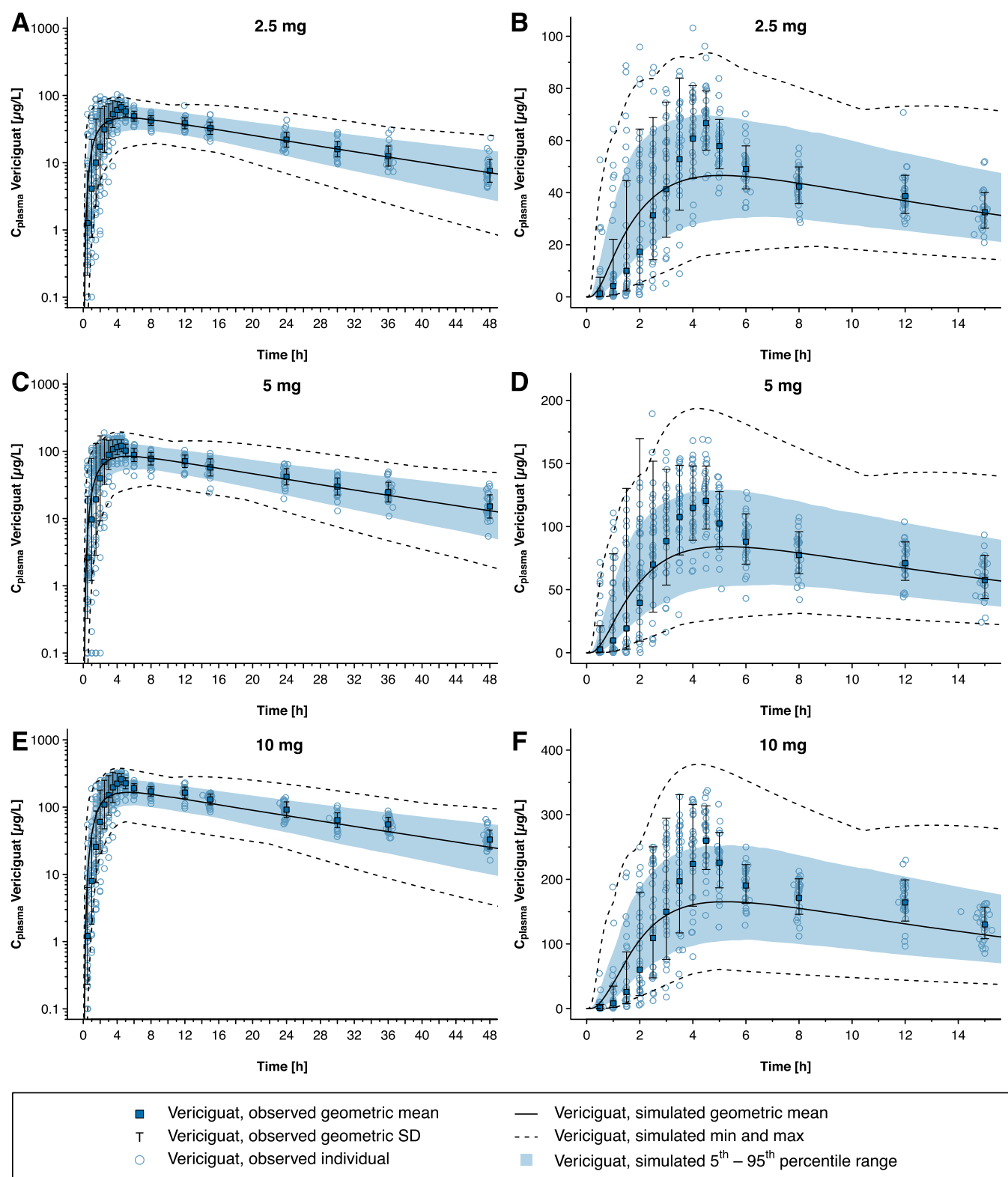


Fig. 7. Simulated *versus* observed plasma concentration-time profiles after dosing a 2.5 mg (A, B), 5 mg (C, D) or 10 mg (E, F) vericiguat tablet following a high-fat, high-caloric meal. The left panels display the first 48 hours with vericiguat concentration on the y-axis using a log scale, right panels show the first 15 hours with vericiguat concentration on the y-axis using a linear scale.

significant inter-occasion variability. While these influences are not fully captured in the current model, mechanistic models could be developed to account for these processes. Such models would allow for the inclusion of inter-occasion variability as well as population differences when

these factors play a critical role in drug absorption.

The simplified dilution model was insufficient to describe gastric pH dynamics following an high-fat, high-caloric meal, likely due to the buffering effects of food components (Simonian et al., 2005). Instead, an

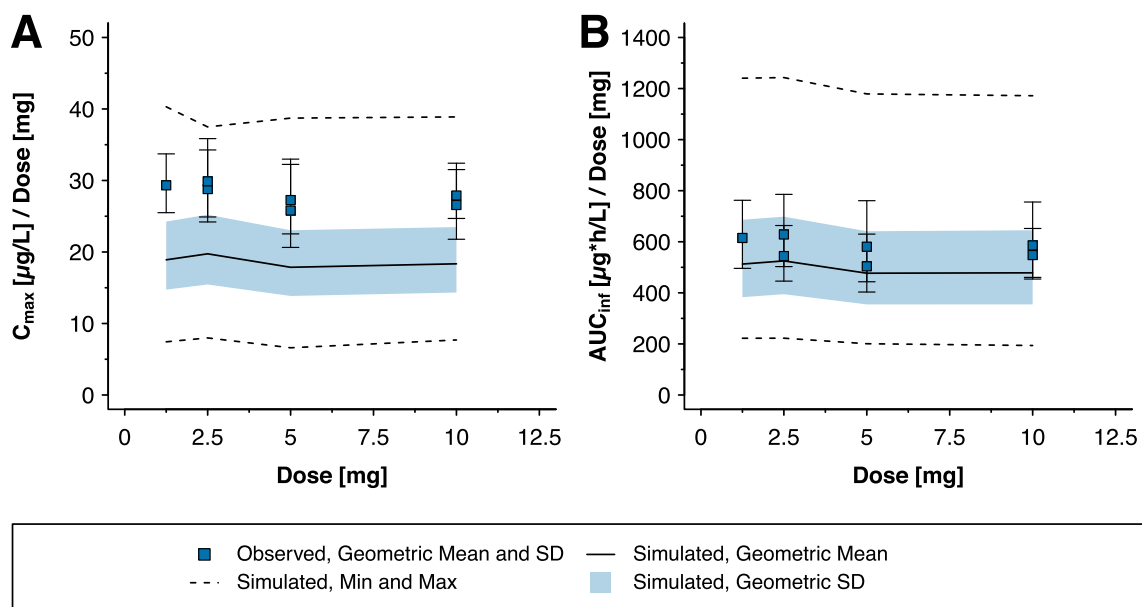


Fig. 8. Observed versus predicted dose normalized pharmacokinetic parameters C_{max} (A) and AUC_{inf} (B), after vericiguat administration following a high-fat, high-caloric meal.

exponential decay function, already available in PK-Sim®, was refitted to improve pH predictions under these fed conditions (Fig. 1B). This adaptation allowed for a better alignment with observed data from studies using telemetric capsules (Dressman et al., 1990; Koziol et al., 2015). However, the data from an aspiration study presented a different profile, possibly due to differences in meal preparation, as the food was homogenized and delivered directly into the stomach through a tube, bypassing normal ingestion and initial digestion processes (Malagelada et al., 1979; Pentafragka et al., 2020). To accurately model the return to basal fasted-state pH, adjustments had to be made beyond simple volume-related mechanisms. Due to slower meal emptying in PK-Sim® (where drug and food emptying are decoupled) and the increase in gastric acid secretion seen *in vivo*, gastric pH in the model declines faster than meal emptying alone would predict (Kiyota et al., 2022; Winter et al., 2023). This approach may present limitations in cases where a fed administration is followed by fasted-state dosing, as the model remains insensitive to water intake following a meal event.

After meal intake, an empirical quadratic function was applied to model the decrease in upper intestinal pH and its gradual return to baseline. This approach effectively captured postprandial pH dynamics observed following a standard FDA meal (Pentafragka et al., 2020). For luminal bile salt concentration dynamics, an empirical linear function was implemented to describe the decline in bile salt concentration after a meal. This decrease was assumed to be similar in the duodenum and upper jejunum, while the lower jejunum remained unaffected, assumed due to a lack of data in this luminal segment. This indicates that the simplified model has limitations, and a more mechanistic approach is recommended for future work to capture bile salt variability more accurately, as suggested by Stamatoopoulos et al. (2020). These authors incorporated cyclical fasted and fed state bile salt concentrations through the interplay between gall bladder volume and the interdigestive migrating motor complex cycle (Stamatoopoulos et al., 2020).

The intraluminal hydrodynamic model implemented in PK-Sim® is based on several assumptions to simplify the complex *in vivo* hydrodynamics of the GI tract. Firstly, it assumed that the vericiguat particles move synchronously with the intestinal fluid. This assumption is likely valid for the small particles used in the formulation ($r_0 < 50 \mu m$). However, for larger or denser particles, the relative flow velocity may need to be considered more carefully. Secondly, the model represents the intestines as a perfect tube with laminar flow, neglecting features

such as intestinal folding and peristaltic movements (Sugano, 2010). These factors could lead to the formation of transient micro-eddies, which may need to be considered in future refinements of the model. While there are many uncertainties in modeling GI hydrodynamics, computational fluid dynamics (CFD) models for the gastric compartment exist and were used to estimate an average value for the fluid velocity $\bar{V}_{fluid}$. However, similar CFD models for the intestines are not readily available. As a result, the PK-Sim® intestinal fluid transfer rates were used to inform the estimation of $\bar{V}_{fluid}$ for the intestines. These simplifications were made due to the computational intensity of CFD models, which makes it difficult to couple them with a PBPK model. Nevertheless, as more mechanistic insights become available in the future, CFD models could potentially provide more accurate representations of GI hydrodynamics and inform future improvements in the model.

4.2. PBBM application

4.2.1. Reparameterization of intestinal permeability

The intestinal permeability of the original PBPK model for vericiguat was adjusted in the new PBBM using data from a SAD clinical study with an oral solution in humans. This enabled an adequate description of PK across a dose range of 0.5–10 mg by the PBBM. However, at the highest 15 mg dose, C_{max} was notably underpredicted (Figure S1). In the original PBPK model, intestinal solubility was optimized using data from *in vivo* clinical studies. In contrast, the bottom-up PBBM approach estimated intestinal solubility of vericiguat based on solubility data in FaSSiF and FeSSiF. Separation of the effect of pH and bile salt micelles on API solubility has the added benefit that it avoids overprediction of solubility in luminal segments where the bile salt concentration is low (e.g., stomach and large intestines), which would be the case if FaSSiF or FeSSiF would be used as reference solubility in the model for solubility-pH profile calculations. The underprediction of the 15 mg dose by the new PBBM model may be due to an underestimation of luminal solubility, as solubility in biorelevant media may not fully reflect actual solubility in the human intestine under fasted or fed conditions. In the absence of direct solubility measurements in human intestinal fluids, applying established correlations between biorelevant media and human intestinal solubility could help address such discrepancies in future modeling efforts.

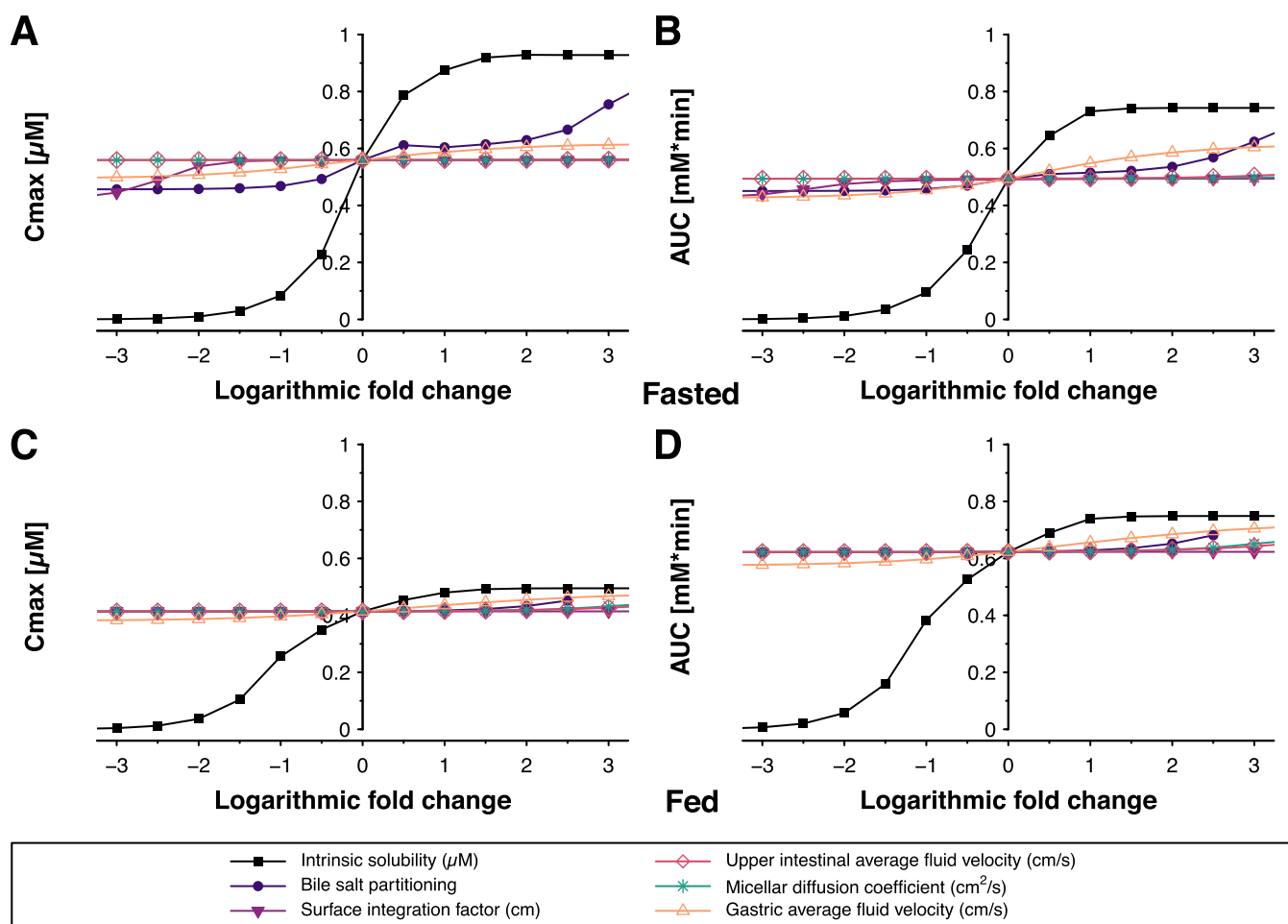


Fig. 9. Sensitivity analysis of absorption related parameters on simulations following administration of a 10 mg vericiguat tablet in the fasted (A-B) and fed (C-D) state. The different shapes represent the effect of adjusting each absorption-related parameter—up to three orders of magnitude above and below the baseline—on the pharmacokinetic parameters C_{max} (A & C) and AUC (B & D). The analyzed parameters include both API and GI tract properties: intrinsic solubility (black squares), neutral and ionized bile salt partitioning constants (deep purple circles), surface integration factor (Magenta downward triangles), upper intestinal average fluid velocity (red diamonds, through an added scaling factor), micellar diffusion coefficient (green stars) and gastric average fluid velocity (orange upward triangles).

(Augustijns et al., 2014). Another potential explanation could be related to how the API fraction available for permeation into enterocytes is handled by the model, especially in scenarios in which a supersaturated state in the intestinal lumen is reached and API concentration exceeds the solubility. In the current set-up this fraction was not available for absorption. However, enabling the “supersaturation” function in PK-Sim®, which allows the API fraction exceeding solubility to be available for absorption, did not substantially improve model predictions across doses. This was likely because, under this setting, the fitted permeability value remained relatively low, resulting in under-prediction of vericiguat pharmacokinetics at lower doses. Finally, the small clinical sample size in this dosing group ($n = 4$), half the size of the other dosing groups, likely affects the reliability of the observed data; by contrast, an additional 21 subjects (study B and D) were included in the 5 mg group. Given the context of use, the estimated intestinal permeability was considered adequate.

To extend the PBBM workflow’s bottom-up capabilities further in the future, it would be beneficial to develop tools that can mechanistically translate permeability data from experimental methods (e.g., PAMPA or Caco-2 assays) into a specific intestinal permeability that can be used in the base PBPK model. This would require updates to the current permeability model, particularly to account for the unstirred water layer (UWL) that forms a diffusion barrier around mucosal tissue. This layer

can vary in thickness across API species, affecting permeability rates (Sugano, 2009b). Such a permeability model would require determination of the extent to which each API species (free drug, micelle bound drug and nanoparticles) diffuses into the UWL and contributes to absorption (Yang et al., 2024). This could enhance the final PBBM’s ability to predict bioavailability by processes that affect colloidal structures, such as differences in bile salt micelles between the fasted and fed states, the presence of surfactants from the formulation, and the formation of nanoparticles. These factors can significantly influence diffusion into the UWL, either facilitating or hindering API transport depending on the physicochemical properties of the drug and formulation.

4.2.2. Fasted state simulations

Integration of the *in vitro* informed P-PSDs to model the tablet dissolution in the PBBM gave a good description of the vericiguat PK in the fasted state (Fig. 2 & Fig. 3). Simulated C_{max} and AUC were close to the clinically observed values and no pronounced non-linearity was predicted or observed over the investigated dose range (Fig. 4). Interestingly, there was no reduction in the dose normalized PK parameters observed between the 1.25 mg and the 15 mg (as three 5 mg tablets) administrations, while a slight reduction was predicted. Predicted t_{max} in the fasted state fell within the clinically observed range, but rather at

the lower end. Again, this could be related to the permeability model and whether micelle bound API is considered available for absorption.

The sensitivity analysis highlights intrinsic solubility as the primary parameter influencing predictions of $C_{\max}$ and AUC, with changes ranging from 48 % to 85 % following a ± 1 log unit change in parameter value. Bile salt partitioning was identified as the second most impactful factor for vericiguat in the fasted state, with corresponding changes of 4 % to 16 % under the same log-fold variation (Fig. 9). This finding enhances confidence in the PBBM's ability to capture vericiguat's PK profile, especially as these parameters were derived from *in vitro* data (Vrenken et al., 2025). Additionally, Fig. 9 shows that only gastric $\bar{V}_{\text{fluid}}$ influenced vericiguat's PK parameters, 4–8 % in $C_{\max}$ and AUC after ± 1 log fold change. In contrast, upper intestinal hydrodynamics had no measurable impact, suggesting that dissolution in the stomach is the primary determinant of the PK outcome. This aligns with vericiguat's classification as a BCS Class II compound, where solubility, rather than permeability, is the limiting factor (Becker et al., 2022).

4.2.3. PPI treatment simulations

The effect of PPI pretreatment, 40 mg omeprazole once daily for 5 days, was modeled by adjusting the mean gastric pH and its inter-individual variability in the population. This adjustment proved effective to predict the decrease in $C_{\max}$ and AUC_{inf} quantitatively. However, the underprediction of $t_{\max}$ was even more pronounced during these simulations (Fig. 5). This underprediction could be related to delayed gastric emptying following PPI treatment, although reports in the literature are conflicting on the effect on gastric emptying in the fasted state (Sanaka et al., 2010). Another reason could be related to vericiguat's predicted pKa value (4.68), which is very close to the mean pH value of 4.6 extracted from the literature, making this a very sensitive parameter in this pH range (Kirchheiner et al., 2009). Interestingly, the plasma concentration-time profile in the control arm of the clinical study seems to be delayed compared to the profile observed in the same dosing scenario of a different study (Fig. 3D) which could point to some individuals in the study group exhibiting a higher than expected $t_{\max}$, even in absence of omeprazole.

4.2.4. Fed state simulations

Fig. 6 and Fig. 7 show that the PBBM effectively captures the delayed $t_{\max}$ of vericiguat in the fed state without requiring any additional parameter adjustments, other than the implemented meal-related changes (being increases in gastric volume, emptying time, luminal pH and bile salt concentrations). The AUC was also reasonably well predicted, though $C_{\max}$ was consistently underpredicted in the fed state (Fig. 9). Sensitivity analysis revealed that $C_{\max}$ was only slightly affected, even when key parameters were increased by several orders of magnitude. This suggests a mechanism influencing $C_{\max}$ that is not currently accounted for in the model.

Examining individual plasma profiles (Fig. 6 & Fig. 7) revealed two distinct patterns: one group showed a delayed appearance of vericiguat in plasma, up to two hours, while the other group displayed a much faster onset. This discrepancy may stem from factors affecting tablet disintegration in the fed state, or possibly from the “Magenstrasse” effect, or “stomach road” (Grimm et al., 2017). This phenomenon occurs when water ingested in a fed state can move rapidly through the stomach, carrying dissolved API or small particles that enable an early plasma appearance. This is different from typical gastric emptying, where the API exits the stomach with the meal.

A similar effect was observed in a clinical study of febuxostat tablets (Adenuric®) and was attributed to variable gastric pressure events. These events were successfully replicated using the GastroDuo dissolution system (Winter et al., 2024). Testing vericiguat tablets with this system could help determine if the formulation is sensitive to these pressure events, which could then inform model adjustments. Integrating these mechanisms into the PBBM may improve $C_{\max}$ predictions

in the fed state. Although the current model structure does not directly replicate this physiological phenomenon, sensitivity to postprandial gastric emptying was investigated by adjusting the β (beta) parameter in the Weibull function used by PK-Sim® to describe drug emptying from the stomach after a meal. A 3-fold increase in this parameter resulted in slow initial drug emptying, followed by a fast release phase, ultimately improving the description of $C_{\max}$ for the vericiguat 10 mg dose (FE: 0.99, Figure S2). This warrants further investigation into the implemented gastric emptying model.

The predicted *versus* observed vericiguat profiles showed that the PBBM effectively captured the observed range of values, as evidenced by the predicted minimum and maximum profiles (Figs. 2–8). However, the observed variability was not fully captured within the 90 % confidence intervals in either the fasted or fed states, with the discrepancy being most pronounced at higher doses. This discrepancy could arise from the current implementation in PK-Sim®, where intraluminal pH and bile salt concentrations, along with population variability, are generated independently when constructing the population. This approach can result in unrealistic individual profiles, such as combinations of low duodenal and high upper jejunal bile salt concentrations. To address this, a more mechanistic implementation of physiological parameters could be considered, ensuring that pH and bile salt concentrations vary in a correlated manner across luminal segments. Alternatively, establishing inter-parameter correlations would help maintain realistic profiles within the population, potentially enhancing the model's ability to capture vericiguat variability at higher doses.

5. Concluding remarks

This publication introduces a novel, fully free and open-source PBBM workflow developed within the OSP ecosystem. The workflow integrates solubility and dissolution parameters, derived from *in vitro* experiments, into the PK-Sim® PBPK modeling platform, extending it into a PBBM framework to predict *in vivo* pharmacokinetics following oral dosing. Leveraging three open-source tools, it enables the characterization of API solubility, formulation dissolution, and *in vivo* absorption. The approach was successfully applied to vericiguat, accurately predicting its pharmacokinetics in fasted, fed, and PPI-treated states. While the *in vivo* predictions captured vericiguat's pharmacokinetics reasonably well, especially considering the bottom-up nature of the workflow, there are still opportunities for refinement, particularly in enhancing the mechanistic descriptions of permeability and luminal bile salt dynamics. Despite these areas for improvement, the workflow provides a solid foundation for the OSP community and is readily extensible to broader PBBM applications. With further validation through additional case studies, it holds strong potential to support biopharmaceutical risk assessment, formulation bridging, dose selection, and *in silico* predictions of bioavailability and bioequivalence. The complete workflow is publicly available on OSP GitHub (<https://github.com/Open-Systems-Pharmacology/Oral-PBBM-Workflow>).

CRedit authorship contribution statement

Paul Vrenken: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maria Vertzoni:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Sebastian Frechen:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Juri Solodenko:** Writing – review & editing, Software. **Michaela Meyer:** Writing – review & editing. **Uwe Muenster:** Writing – review & editing. **André Dallmann:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

All authors declare no competing interests for this work. P.V., A.D., U.M., J.S., M.M., and S.F. are/were employees and potential stockholders of Bayer AG and may own stock in the company.

J.S. is a member of the OSP Management Team.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ejps.2025.107189](https://doi.org/10.1016/j.ejps.2025.107189).

Data availability

The complete PBBM workflow and modeling files are available on OSP GitHub (github.com/Open-Systems-Pharmacology/Oral-PBBM-Workflow).

References

- Augustijns, P., Wuyts, B., Hens, B., Annaert, P., Butler, J., Brouwers, J., 2014. A review of drug solubility in human intestinal fluids: implications for the prediction of oral absorption. In: European Journal of Pharmaceutical Sciences, Special Issue on 7th International Symposium on Microdialysis - Edited By: William Couet and Hartmut Derendorf & Special Issue on The status of understanding and predicting gastrointestinal drug absorption and the way forward by means of the novel EU project OrBiTo - Edited By: Bertil Abrahamsson, Peter Langguth, Hans Lennernäs, 57, pp. 322–332. <https://doi.org/10.1016/j.ejps.2013.08.027>.
- Avdeef, A., Nielsen, P.E., Tsinman, O., 2004. PAMPA—A drug absorption *in vitro* model: 11. Matching the *in vivo* unstirred water layer thickness by individual-well stirring in microtitre plates. Eur. J. Pharm. Sci. 22, 365–374. <https://doi.org/10.1016/j.ejps.2004.04.009>.
- Bayer - Clinical Trials Explorer, 2025. [WWW Document]. URL: <https://clinicaltrials.bayer.com/studies/> (accessed 12.12.24).
- Becker, C., Boettcher, M., Muenster, U., Loewen, S., Lobmeyer, M., Mueck, W., 2022. Results from *in vitro* and *in vivo* studies evaluating the bioavailability, effects of food, and administration as crushed tablet suspension on vericiguat pharmacokinetics. AAPS. Open. <https://doi.org/10.1186/s41120-022-00063-4>.
- Boettcher, M., Gerisch, M., Lobmeyer, M., Besche, N., Thomas, D., Gerrits, M., Lemmen, J., Mueck, W., Radtke, M., Becker, C., 2020. Metabolism and pharmacokinetic drug-drug interaction profile of Vericiguat, A soluble guanylate cyclase stimulator: results from preclinical and phase I healthy volunteer studies. Clin. Pharmacokinet. <https://doi.org/10.1007/s40262-020-00895-x>.
- Brake, K., Gumireddy, A., Tiwari, A., Chauhan, H., Kumari, D., 2017. *In vivo* studies for drug development via oral delivery: challenges, animal models and techniques. Pharm. Anal. Acta 08. <https://doi.org/10.4172/2153-2435.1000560>.
- Dallmann, A., Teutonico, D., Schaller, S., Burghaus, R., Frechen, S., 2024. In-depth analysis of the selection of PBPK modeling tools: bibliometric and social network analysis of the open systems pharmacology community. J. Clin. Pharmacol. <https://doi.org/10.1002/jcph.2453>.
- Chang, W., Cheng, J., Allaire, J., Sievert, C., Schloerke, B., Xie, Y., McPherson, J., Davis, T., Ushey, K., Allen, J., Dipert, A., Borges, B., shiny: Web Application Framework for R, Posit Software, PBC, Boston, MA, 2024, version 1.8.1.1. URL: <https://shiny.posit.co/>.
- Dressman, J.B., Berardi, R.R., Dermentzoglou, L.C., Russell, T.L., Schmaltz, S.P., Barnett, J.L., Jarvenpaa, K.M., 1990. Upper gastrointestinal (GI) pH in young, healthy men and women. Pharm. Res. <https://doi.org/10.1023/a:1015827908309>.
- FDA, 2021. FDA label VERQUVO (vericiguat).
- FDA, 2020. The use of physiologically based pharmacokinetic analyses — biopharmaceutics applications for oral drug product development. Draft. Guid.
- Frechen, S., Ince, I., Dallmann, A., Gerisch, M., Jungmann, N.A., Becker, C., Lobmeyer, M., Trujillo, M.E., Xu, S., Burghaus, R., Meyer, M., 2023. Applied physiologically-based pharmacokinetic modeling to assess uridine diphosphate-glucuronosyltransferase-mediated drug-drug interactions for Vericiguat. CPT. Pharmacomet. Syst. Pharmacol. <https://doi.org/10.1002/psp4.13059>.
- Gamsiz, E.D., Ashtikar, M., Crison, J., Wolosz, W., Bolger, M.B., Carrier, R.L., 2010. Predicting the effect of fed-state intestinal contents on drug dissolution. Pharm. Res. <https://doi.org/10.1007/s11095-010-0264-x>.
- Grimm, M., Scholz, E., Koziolk, M., Kühn, J.-P., Weitschies, W., 2017. Gastric water emptying under Fed State clinical trial conditions is as fast as under fasted conditions. Mol. Pharm. 14, 4262–4271. <https://doi.org/10.1021/acs.molpharmaceut.7b00623>.
- Heimbach, T., Suarez-Sharp, S., Kakhi, M., Holmstock, N., Olivares-Morales, A., Pepin, X., Sjogren, E., Tsakalozou, E., Seo, P., Li, M., Zhang, X., Lin, H.P., Montague, T., Mitra, A., Morris, D., Patel, N., Kesiosoglou, F., 2019. Dissolution and translational modeling strategies toward establishing an *In Vitro-In vivo* link-a workshop summary report. AAPS. J. <https://doi.org/10.1208/s12248-019-0298-x>.
- Hens, B., Seegobin, N., Bermejo, M., Tsume, Y., Clear, N., McAllister, M., Amidon, G.E., Amidon, G.L., 2022. Dissolution challenges associated with the surface pH of drug particles: integration into mechanistic oral absorption modeling. AAPS. J. <https://doi.org/10.1208/s12248-021-00663-0>.
- Hummler, H., Sarwinska, D., Weitschies, W., Gollasch, M., Page, S., 2023. Parameters to consider for successful medication use in older adults - an AGEPOP review. Eur. J. Pharm. Sci. 187, 106453. <https://doi.org/10.1016/j.ejps.2023.106453>.
- Kambayashi, A., Dressman, J.B., 2022. Towards virtual bioequivalence studies for oral dosage forms containing poorly water-soluble drugs: a physiologically based biopharmaceutics modeling (PBBM) approach. J. Pharm. Sci. <https://doi.org/10.1016/j.xphs.2021.08.008>.
- Kirchheiner, J., Glatt, S., Fuhr, U., Klotz, U., Meineke, I., Seufferlein, T., Brockmoller, J., 2009. Relative potency of proton-pump inhibitors-comparison of effects on intragastric pH. Eur. J. Clin. Pharmacol. <https://doi.org/10.1007/s00228-008-0576-5>.
- Kiyota, T., Kambayashi, A., Takagi, T., Yamashita, S., 2022. Importance of Gastric Secretion and the Rapid Gastric Emptying of Ingested Water along the Lesser Curvature ("Magenstrasse") in Predicting the *In Vivo* Performance of Liquid Oral Dosage Forms in the Fed State Using a Modeling and Simulation. Mol. Pharm. <https://doi.org/10.1021/acs.molpharmaceut.1c00778>.
- Koenigsnecht, M.J., Baker, J.R., Wen, B., Frances, A., Zhang, H., Yu, A., Zhao, T., Tsume, Y., Pai, M.P., Bleske, B.E., Zhang, X., Lionberger, R., Lee, A., Amidon, G.L., Hasler, W.L., Sun, D., 2017. *In vivo* dissolution and systemic absorption of immediate release ibuprofen in Human gastrointestinal tract under fed and fasted conditions. Mol. Pharm. <https://doi.org/10.1021/acs.molpharmaceut.7b00425>.
- Kostewicz, E.S., Wunderlich, M., Brauns, U., Becker, R., Bock, T., Dressman, J.B., 2004. Predicting the precipitation of poorly soluble weak bases upon entry in the small intestine. J. Pharm. Pharmacol. <https://doi.org/10.1211/0022357022511>.
- Kourentas, A., Vertzoni, M., Stavrinoudakis, N., Symillidis, A., Brouwers, J., Augustijns, P., Reppas, C., Symillides, M., 2016. An *in vitro* biorelevant gastrointestinal transfer (BioGIT) system for forecasting concentrations in the fasted upper small intestine: design, implementation, and evaluation. Eur. J. Pharm. Sci. <https://doi.org/10.1016/j.ejps.2015.11.012>.
- Koziolek, M., Schneider, F., Grimm, M., Modebeta, C., Seekamp, A., Roustom, T., Siegmund, W., Weitschies, W., 2015. Intragastric pH and pressure profiles after intake of the high-caloric, high-fat meal as used for food effect studies. J. Control. Release. <https://doi.org/10.1016/j.jconrel.2015.10.022>.
- Li, Y., Kong, F., 2022. Simulating human gastrointestinal motility in dynamic *in vitro* models. Compr. Rev. Food. Sci. Food. Saf. 21, 3804–3833. <https://doi.org/10.1111/1541-4337.13007>.
- Lindahl, A., Ungell, A.-L., Knutson, L., Lennernäs, H., 1997. Characterization of fluids from the stomach and proximal jejunum in men and women. Pharm. Res 14, 497–502. <https://doi.org/10.1023/A:1012107801889>.
- Lindfors, L., Forssen, S., Westergren, J., Olsson, U., 2008. Nucleation and crystal growth in supersaturated solutions of a model drug. J. Colloid. Interface. Sci. <https://doi.org/10.1016/j.jcis.2008.05.034>.
- Lippert, J., Burghaus, R., Edginton, A., Frechen, S., Karlsson, M., Kovar, A., Lehr, T., Milligan, P., Nock, V., Ramusovic, S., Riggs, M., Schaller, S., Schlender, J., Schmidt, S., Sevestre, M., Sjogren, E., Solodenko, J., Staab, A., Teutonico, D., 2019. Open systems pharmacology community—an open access, open source, open science approach to modeling and simulation in pharmaceutical sciences. CPT. Pharmacomet. Syst. Pharmacol. <https://doi.org/10.1002/psp4.12473>.
- Litou, C., Vertzoni, M., Goumas, C., Vasdekis, V., Xu, W., Kesiosoglou, F., Reppas, C., 2016. Characteristics of the human upper gastrointestinal contents in the fasted state under hypo- and A-chlorhydric gastric conditions under conditions of typical drug-drug interaction studies. Pharm. Res. <https://doi.org/10.1007/s11095-016-1882-8>.
- Luo, D., Wan, X., Liu, J., Tong, T., 2018. Optimally estimating the sample mean from the sample size, median, mid-range, and/or mid-quartile range. Stat. Methods. Med. Res. <https://doi.org/10.1177/0962280216669183>.
- Mackie, C., Arora, S., Seo, P., Moody, R., Rege, B., Pepin, X., Heimbach, T., Tannergren, C., Mitra, A., Suarez-Sharp, S., Borges, L.N., Kijima, S., Kotzagiorgis, E., Malamataris, M., Veerasingham, S., Polli, J.E., Rullo, G., 2024. Physiologically based Biopharmaceutics modeling (PBBM): best practices for drug product quality, regulatory and industry perspectives: 2023 workshop summary report. Mol. Pharm. <https://doi.org/10.1021/acs.molpharmaceut.4c00202>.
- Malagelada, J.-R., Go, V.L.W., Summerskill, W.H.J., 1979. Different gastric, pancreatic, and biliary responses to solid-liquid or homogenized meals. Dig. Dis. Sci. 24, 101–110. <https://doi.org/10.1007/BF01324736>.
- Mann, J., Dressman, J., Rosenblatt, K., Ashworth, L., Muenster, U., Frank, K., Hutchins, P., Williams, J., Klumpp, L., Wielockx, K., Berben, P., Augustijns, P., Holm, R., Hofmann, M., Patel, S., Beato, S., Ojala, K., Tomaszewska, I., Bruel, J.L., Butler, J., 2017. Validation of dissolution testing with biorelevant Media: an OrBiTo study. Mol. Pharm. <https://doi.org/10.1021/acs.molpharmaceut.7b00198>.
- Manolis, E., Garcia-Arieta, A., Lindahl, A., Kotzagiorgis, E., Limberg, J., Holte, O., Paixao, P., Versantvoort, C., Tshinanu, F.M., Blake, K., Van Den Heuvel, M., 2023. Using mechanistic models to support development of complex generic drug products: european medicines agency perspective. CPT. Pharmacomet. Syst. Pharmacol. <https://doi.org/10.1002/psp4.12906>.
- Markham, A., Duggan, S., 2021. Vericiguat: first approval. Drugs. <https://doi.org/10.1007/s40265-021-01496-z>.

- Mullertz, A., Reppas, C., Psachoulas, D., Vertzoni, M., Fatouros, D.G., 2015. Structural features of colloidal species in the human fasted upper small intestine. *J. Pharm. Pharmacol.* <https://doi.org/10.1111/jphp.12336>.
- Musther, H., Olivares-Morales, A., Hatley, O.J.D., Liu, B., Rostami-Hodjegan, A., 2014. Animal versus human oral drug bioavailability: do they correlate? Special Issue on 7th International Symposium on Microdialysis - Edited By: William Couet and Hartmut Derendorf & Special Issue on The status of understanding and predicting gastrointestinal drug absorption and the way forward by means of the novel EU project OrBiTo - Edited By: Bertil Abrahamsson, Peter Langguth, Hans Lennernäs. *Europ. J. Pharmacol. Sci.* 57, 280–291. <https://doi.org/10.1016/j.ejps.2013.08.018>.
- Okazaki, A., Mano, T., Sugano, K., 2008. Theoretical dissolution model of poly-disperse drug particles in biorelevant media. *J. Pharm. Sci.* <https://doi.org/10.1002/jps.21070>.
- OSP: Oral-PBBM-Workflow, 2025. [WWW Document]. URL: <https://github.com/Open-Systems-Pharmacology/Oral-PBBM-Workflow> (accessed 11.26.24).
- Pathak, S.M., Ruff, A., Kostewicz, E.S., Patel, N., Turner, D.B., Jamei, M., 2017. Model-based analysis of biopharmaceutic experiments to improve mechanistic oral absorption modeling: an integrated *in vitro in vivo* extrapolation perspective using ketoconazole as a Model drug. *Mol. Pharm.* <https://doi.org/10.1021/acs.molpharmaceut.7b00406>.
- Pentafragka, C., Vertzoni, M., Dressman, J., Symillides, M., Goumas, K., Reppas, C., 2020. Characteristics of contents in the upper gastrointestinal lumen after a standard high-calorie high-fat meal and implications for the *in vitro* drug product performance testing conditions. *Eur. J. Pharm. Sci.* <https://doi.org/10.1016/j.ejps.2020.105535>.
- Pepin, X., Goetschy, M., Abrahamsen-Alami, S., 2022. Mechanistic models for USP2 dissolution apparatus, including fluid hydrodynamics and sedimentation. *J. Pharm. Sci.* <https://doi.org/10.1016/j.xphs.2021.10.006>.
- Pepin, X.J., Flanagan, T.R., Holt, D.J., Eidelman, A., Treacy, D., Rowlings, C.E., 2016. Justification of drug product dissolution rate and drug substance particle size specifications based on absorption PBPK modeling for Lesinurad Immediate Release tablets. *Mol. Pharm.* <https://doi.org/10.1021/acs.molpharmaceut.6b00497>.
- Pepin, X.J.H., Moir, A.J., Mann, J.C., Sanderson, N.J., Barker, R., Meehan, E., Plumb, A.P., Bailey, G.R., Murphy, D.S., Krejsa, C.M., Andrew, M.A., Ingallinera, T.G., Slater, J.G., 2019a. Bridging *in vitro* dissolution and *in vivo* exposure for acalabrutinib. Part II. A mechanistic PBPK model for IR formulation comparison, proton pump inhibitor drug interactions, and administration with acidic juices. *Eur. J. Pharm. Biopharm.* <https://doi.org/10.1016/j.ejpb.2019.07.011>.
- Pepin, X.J.H., Sanderson, N.J., Blanz, A., Grover, S., Ingallinera, T.G., Mann, J.C., 2019b. Bridging *in vitro* dissolution and *in vivo* exposure for acalabrutinib. Part I. Mechanistic modelling of drug product dissolution to derive a P-PSD for PBPK model input. *Eur. J. Pharm. Biopharm.* <https://doi.org/10.1016/j.ejpb.2019.07.014>.
- Petrakis, O., Vertzoni, M., Angelou, A., Kesiosoglou, F., Bentz, K., Goumas, K., Reppas, C., 2015. Identification of key factors affecting the oral absorption of salts of lipophilic weak acids: a case example. *J. Pharm. Pharmacol.* <https://doi.org/10.1111/jphp.12320>.
- Pohl, P., Saparov, S.M., Antonenko, Y.N., 1998. The size of the unstirred layer as a function of the solute diffusion coefficient. *Biophys. J.* [https://doi.org/10.1016/s0006-3495\(98\)74058-5](https://doi.org/10.1016/s0006-3495(98)74058-5).
- Reppas, C., Kuentz, M., Bauer-Brandl, A., Carlet, S., Dallmann, A., Dietrich, S., Dressman, J., Ejekjaer, L., Frechen, S., Guidetti, M., Holm, R., Holzem, F.L., Karlsson, E., Kostewicz, E., Paulus, F., Senniksen, M.B., Stillhart, C., Turner, D.B., Vertzoni, M., Vrenken, P., Zoller, L., Griffin, B.T., O'Dwyer, P.J., 2023. Leveraging the use of *in vitro* and computational methods to support the development of enabling oral drug products: an InPharma commentary. *Eur. J. Pharm. Sci.* <https://doi.org/10.1016/j.ejps.2023.106505>.
- Rivera, K.R., Pessi, J., Andersson, V., Gustafsson, H., Gluud, L.L., Buckley, S.T., 2023. Characterizing interspecies differences in gastric fluid properties to improve understanding of *in vivo* oral drug formulation performance. *Eur. J. Pharm. Sci.* 183, 106386. <https://doi.org/10.1016/j.ejps.2023.106386>.
- Rohatgi, A., 2024. WebPlotDigitizer.
- Sanaka, M., Yamamoto, T., Kuyama, Y., 2010. Effects of proton pump inhibitors on gastric emptying: a systematic review. *Dig. Dis. Sci.* 55, 2431–2440. <https://doi.org/10.1007/s10620-009-1076-x>.
- Simonian, H.P., Vo, L., Doma, S., Fisher, R.S., Parkman, H.P., 2005. Regional postprandial differences in pH within the stomach and gastroesophageal junction. *Dig. Dis. Sci.* 50, 2276–2285. <https://doi.org/10.1007/s10620-005-3048-0>.
- Sjögren, E., Westergren, J., Grant, I., Hanisch, G., Lindfors, L., Lennernäs, H., Abrahamsson, B., Tannergren, C., 2013. In silico predictions of gastrointestinal drug absorption in pharmaceutical product development: application of the mechanistic absorption model GI-Sim. *Eur. J. Pharm. Sci.* <https://doi.org/10.1016/j.ejps.2013.05.019>.
- Stamatopoulos, K., Pathak, S.M., Marciani, L., Turner, D.B., 2020. Population-based PBPK model for the prediction of time-variant bile salt disposition within GI luminal fluids. *Mol. Pharm.* <https://doi.org/10.1021/acs.molpharmaceut.0c00019>.
- Sugano, K., 2010. Aqueous boundary layers related to oral absorption of a drug: from dissolution of a drug to carrier mediated transport and intestinal wall metabolism. *Mol. Pharm.* <https://doi.org/10.1021/mp1001119>.
- Sugano, K., 2009a. Introduction to computational oral absorption simulation. *Expert. Opin. Drug. Metab. Toxicol.* <https://doi.org/10.1517/17425250902835506>.
- Sugano, K., 2009b. Estimation of effective intestinal membrane permeability considering bile micelle solubilisation. *Int. J. Pharm.* <https://doi.org/10.1016/j.ijpharm.2008.10.001>.
- Sugano, K., Okazaki, A., Sugimoto, S., Tavornvipsa, S., Omura, A., Mano, T., 2007. Solubility and dissolution profile assessment in drug discovery. *Drug. Metab. Pharmacokinet.* <https://doi.org/10.2133/dmpk.22.225>.
- Thelen, K., Coboecken, K., Willmann, S., Burghaus, R., Dressman, J.B., Lippert, J., 2011. Evolution of a detailed physiological model to simulate the gastrointestinal transit and absorption process in humans, part 1: oral solutions. *J. Pharm. Sci.* <https://doi.org/10.1002/jps.22726>.
- Thelen, K., Coboecken, K., Willmann, S., Dressman, J.B., Lippert, J., 2012. Evolution of a detailed physiological model to simulate the gastrointestinal transit and absorption process in humans, part II: extension to describe performance of solid dosage forms. *J. Pharm. Sci.* <https://doi.org/10.1002/jps.22825>.
- Uekusa, T., Avdeef, A., Sugano, K., 2021. Is equilibrium slurry pH a good surrogate for solid surface pH during drug dissolution? *Eur. J. Pharm. Sci.* <https://doi.org/10.1016/j.ejps.2021.106037>.
- Vrenken, P., Vertzoni, M., Frechen, S., Solodenko, J., Meyer, M., Muenster, U., Dallmann, A., 2025. Development of a novel physiologically based biopharmaceutics modeling (PBBM) framework using the open systems pharmacology suite, part 1: *in vitro* modeling of vericiguat. *Eur. J. Pharm. Sci.* 212, 107164. <https://doi.org/10.1016/j.ejps.2025.107164>.
- Wan, X., Wang, W., Liu, J., Tong, T., 2014. Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC. Med. Res. Methodol.* <https://doi.org/10.1186/1471-2288-14-135>.
- Willmann, S., Thelen, K., Becker, C., Dressman, J.B., Lippert, J., 2010. Mechanism-based prediction of particle size-dependent dissolution and absorption: cilostazol pharmacokinetics in dogs. *Eur. J. Pharm. Biopharm.* <https://doi.org/10.1016/j.ejpb.2010.06.003>.
- Winter, F., Foja, C., Feldmuller, M., Kromrey, M.L., Schick, P., Tzvetkov, M., Weitschies, W., 2024. Predicting gastric emptying of drug substances taken under postprandial conditions by combination of biorelevant dissolution and mechanistic in silico modeling. *Eur. J. Pharm. Sci.* <https://doi.org/10.1016/j.ejps.2024.106788>.
- Winter, F., Schick, P., Weitschies, W., 2023. Bridging the Gap between Food Effects under Clinical Trial Conditions and Real Life: Modeling Delayed Gastric Emptying of Drug Substances and Gastric Content Volume Based on Meal Characteristics. *Mol. Pharm.* <https://doi.org/10.1021/acs.molpharmaceut.2c00782>.
- Wu, D., Li, M., 2022. Current State and challenges of physiologically based biopharmaceutics modeling (PBBM) in oral drug product development. *Pharm. Res.* <https://doi.org/10.1007/s11095-022-03373-0>.
- Yang, D.H., Najafian, S., Chaudhuri, B., Li, N., 2024. The particle drifting effect: a combined function of colloidal and drug properties. *Mol. Pharm.* 21, 5510–5528. <https://doi.org/10.1021/acs.molpharmaceut.4c00751>.